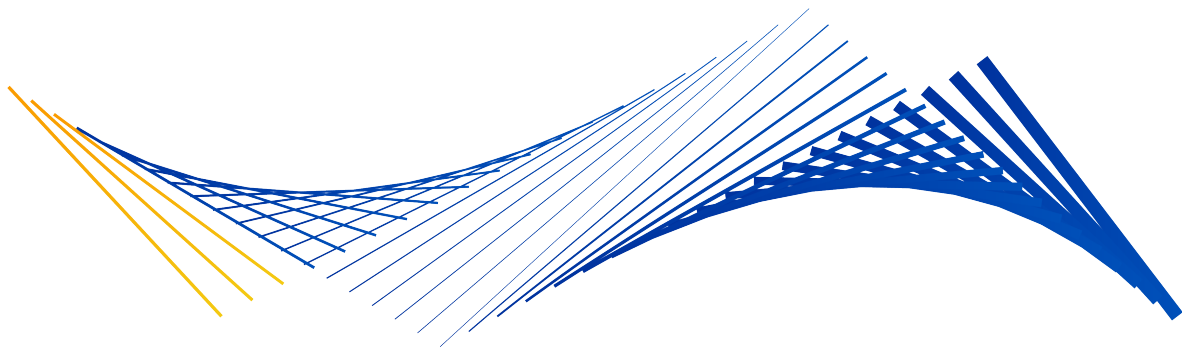


Mobile Push Payment Program Implementation Guide

Effective: 1 June 2022

Version 2.4



Visa Supplemental Requirements

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Chapter 1

Mobile Push Payment Program Overview

About This Guide

This guide is the primary information source for implementation of the Mobile Push Payment solution by Visa clients and partners.

The guide is meant for use by issuer, acquirer, partner, and processor staff who are involved in the deployment of the solution or in managing the ongoing operation of its program.

For issuers, this guide provides information on capability required in processes and systems to deliver the solution to their consumers. These cover mobile banking, card or account management, customer lifecycle management, transaction processing and user interface.

For acquirers, this guide provides information on capability required in processes and systems to deliver the solution to their merchants. These cover merchant lifecycle management, transaction processing and user interface.

Executive Summary

This section gives an overview of the Mobile Push Payment Implementation Guide.

Need for the solution: For any financial services provider, payments is the foundation of their relationship with the customer. Information from payments drives engagement and the services offered. Meanwhile, customer expectations are changing rapidly, influenced by other digital experiences and constant mobile usage. They expect on-the-go, secure, always-on, instant and constant engagement. The mobile app or Unstructured Supplementary Service Data (USSD) is becoming the preferred channel for accessing financial accounts instead of the branch. Financial services providers need to drive traffic to these channels with new customer experiences, starting with payments capability.

Introducing the solution: The Mobile Push Payment solution from Visa is designed to address this gap. Mobile Push Payment is a scalable, interoperable, four-party, account-to-account payment solution adapted for a card-less, digital world. It enables a host of transactions including merchant payment use cases, person-to-person money transfer, cash deposit and cash withdrawal. It enables the digital wallet or mobile banking services from participating clients and partners, to drive new payment experience and use cases.

About the solution: Transactions in the solution are initiated by the consumer on their mobile device and authenticated by their issuer as the first step. The program delivers its value by combining the connectivity provided by mobile phones with "push payment" as the network transaction. On VisaNet, the "push payment" is defined as the Original Credit Transaction (OCT). (Visa's push payment platform is also referred to as Visa Direct). The solution is designed for easy implementation through simple enhancements to the client's payment systems. Since the solution is developed over existing Visa transaction messaging capabilities, it can be integrated faster with the existing processing infrastructure and systems of the issuer and the acquirer. The solution uses existing Visa processes for settlement and disputes, making it seamless to operate with other Visa products without incurring additional overhead.

Implementing the solution: To implement the solution, the client's mobile processing system and transaction processing system have to be modified. Clients can reduce their effort in deploying the solution, by utilizing Visa provided tools like mobile software development kit (SDK) and API based integration to Visa system. The mobile SDK helps in fast and easy integration into the client's mobile banking app, payments app or merchant app. The API integration help clients achieve faster implementation and at lower cost by bypassing the need for changes in their payment switch. Existing ISO connected clients who do not wish to impact their Visa processing systems, or new clients not using legacy ISO connectivity, can choose to integrate with VisaNet using these APIs. This guide provides details of implementation activities that issuers, acquirers, partners or processors undertake to deploy the solution.

In summary: For issuers, implementation of the solution adds powerful payment functionality to their mobile banking and to their cardholder accounts. For acquirers, the solution provides a scalable, low cost, acceptance tool for transformational increase in their acceptance footprint.

Summary of Changes

This is a summary of changes made to this document over version 1.0, dated 15 December 2015.
Principal changes are highlighted in the Guide with change bars.

Version 2.0 Summary of Changes

Document Reorganization

The following sections have been re-organized in this implementation guide and overall document size reduced:

- Business Consideration section giving background and rationale for deploying mVisa has been deleted.
- Merchant Data Specifications (in the information display) has been moved to *Mobile Push Payment - Technical Specifications* Document.
- Implementation planning section has been removed; key content of that section is incorporated throughout the document.
- Minor changes and corrections have been made throughout the document.
- One-time PAN use cases have been moved into a separate document.
- Reference to mVisa brand name has been replaced with Mobile Push Payments. Updated with numeric changes and Visa Rules Committee terminology.

Chapter	Description	Audience	Principal Changes/New Sections
Mobile Push Payment Solution and Network Processing	Provides Visa specifications of the solution	Issuer and acquirer	• International (cross border) transaction processing has been clarified. • Risk management topic has significant updates.
Issuer Processes and Processing System	Describes the processes and the processing capability required by the issuer for the solution	Issuer/ Processor/ Operation	• Key VisaNet message fields and their processing has been described. • Configuration architecture option for issuers has been expanded. • Issuer specific implementation checklist has been introduced.
Acquirer Processes and Processing System	Describes the processes and the processing capability required by the acquirer for the solution	Acquirer/ Processor/ Operation	Key VisaNet message fields and their processing has been described.
Appendix Program Requirement	Provides the operating rules for the program	Issuer/ Acquirer	Changes have been made to some of the clauses.

Version 2.1 Summary of Changes

- mVisa, brand name change has been incorporated in the document and a separate Brand Standards guide has been sunset.
- Visa Claims Resolution (VCR) related changes have been incorporated.
- Agent as a separate category is removed, while maintaining those transactions on the network as-is. Cash-in and cash-out transaction can be performed at merchant location.
- Reference has been made to newer functionality such as alias, where reference to one-time PAN use cases have been removed.
- Minor edits have been made throughout the document.

Version 2.2 Summary of Changes

This document is updated over version 2.1, dated 31 May 2018 to remove the Appendix, “Program Requirement”, since the implementation guide is now added as a Visa Supplement Requirement of the Visa Core Rules and Visa Product and Service Rules.

| Version 2.3 Summary of Changes

- The use of MP Business Application Identifier (BAI) on an AFT transaction for funding Mobile Push Payments transactions has been introduced.
- The Dispute Resolution Process section has been updated to include the dispute process stages.
- A new section on Arbitration or Compliance Liability has been added.

| Version 2.4 Summary of Changes

- Added a note on 8 digit Issuing (ISO) BIN in the following sections:
 - Merchant PAN and Merchant ID
 - Issuing (ISO) BIN Requirement and PAN Creation

Related Publications

This section lists other publications which may contain information relevant to Mobile Push Payments.

Related Publications - Primary

Title	Description
Mobile Push Payment - Technical Specifications	Provides specifications for VisaNet messages and for Merchant Data Encoding (like QR code) utilized in the program.
Mobile Push Payment - Mobile User Experience Guide (Mobile app and USSD)	Provides the user experience on the mobile device deployed by issuers and acquirers participating in the program.
Visa Product and Brand Standards	Provides the branding rules, brand usage and brands specifications for the program.

Related Publications - Topic Specific

Topic	Description
Visa Direct Original Credit Transaction (OCT) - Global Implementation Guide	Person-to-person implementation: Provides specifications and implementation information required to implement person-to-person money transfer transaction.
Account Funding Transaction (AFT) Processing Guide	Use AFT for funding Mobile Push Payment transactions: This document provides information needed for implementing AFT.
API Implementation for issuer and acquirer (Website: developer.visa.com)	Deploying API: Provides API specifications to be used by issuer and acquirer for implementation.
Mobile Push Payment - SDK Integration Guide (for issuer and acquirer) (Website: developer.visa.com)	Deploying SDK: Provides details of SDK as relevant to issuer and acquirer for integrating SDK into mobile app. This guide is available within the downloadable files of the SDK.
Mobile Push Payment Test Cases	Testing ISO VisaNet connectivity: For use by client during VisaNet integration and testing.
Merchant Information Display Guide (QR + USSD)	Deploying merchant information display: Provides requirement and guidelines to be followed by acquirer while deploying merchant information display.
App for Client Testing App	Testing mobile app: Country specific guide, for use by client to test consumer app and merchant QR code before deployment.
Use Cases Description Guide	Deploying new use cases: Provides description of Mobile Push Payment use cases that can be implemented by issuers and acquirers.

Chapter 2

Mobile Push Payment Solution and Network Processing

This section provides a description of the solution covering its features, network specifications and operating rules.

The section provides a description of transactions enabled in the program; the use of personal account numbers (PANs); the operational rules governing the program (branding requirements, operating rules, dispute resolution rules and risk management requirements); and the commercial drivers of the program (settlement, interchange reimbursement fees and pricing). This chapter includes important information about the solution for both issuers and acquirers.

Introduction to Mobile Push Payment Solution

The Mobile Push Payment solution adds interoperable and scalable payments ability to accounts from participating clients.

The value of the solution is derived from the combination of mobile accessed accounts with “push payment” as the underlying network transaction. The solution is a channel (mobile channel) to access the Visa network, with VisaNet providing the route for mobile accessible accounts to transact with

each other. In addition to delivering various merchant payment use cases, the solution also enables person-to-person money transfer, account deposit and account withdrawal. The solution is thus adapted for a card-less (no plastic card), device-less (no POS machine), digital world.

Transactions enabled by the solution are open loop, four-party payments, with Visa performing its usual role of a payment ecosystem enabler. The solution delivers value for all network participants:

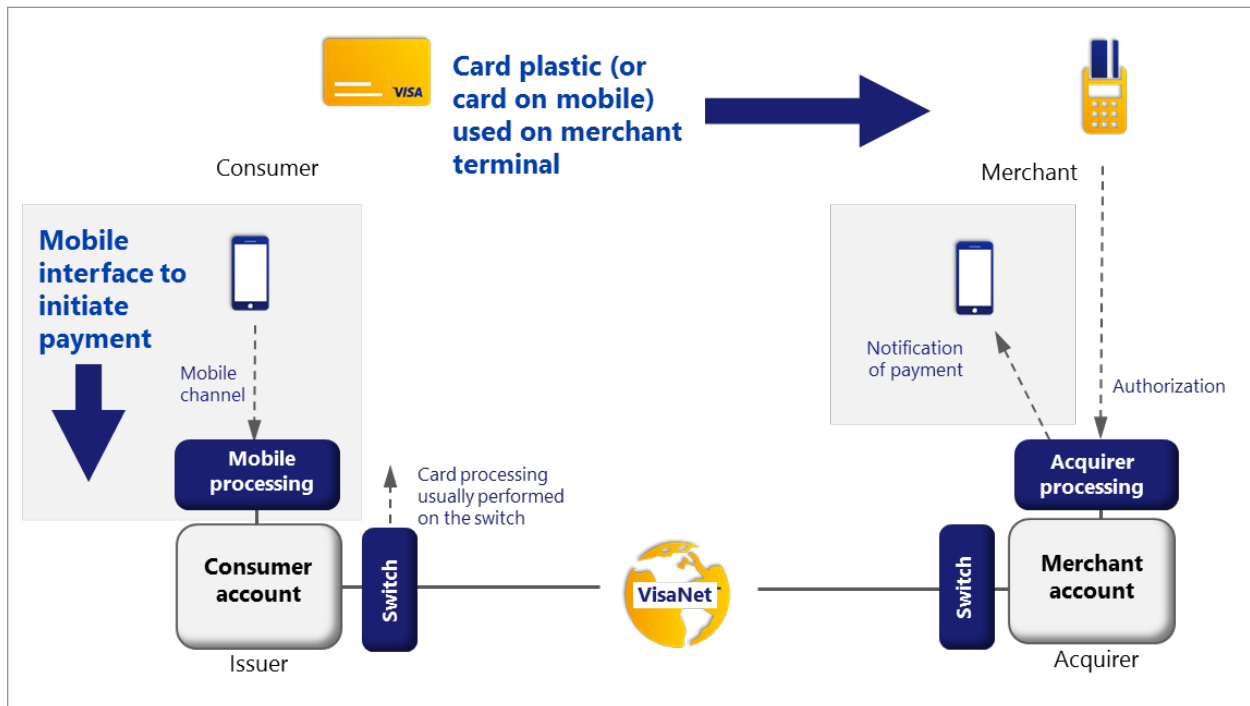
1. For issuers and consumer mobile payment providers, it supplies **versatile transaction ability for their mobile banking or mobile wallet** offering, enabling digital value transfer conducted by a consumer.
2. For acquirers and merchant service providers, it provides **low cost, acquiring capability** to help penetrate the large, unrealized merchant acceptance potential or provide alternate payment experiences at existing merchants.
3. For clients and partners interested in **branchless banking**, it provides options to enable scalable cash deposit and cash withdrawal at merchant location and build a route to digital payments for the mass market.
4. It provides **globally interoperable standards**, which can work on any connected smart mobile devices with fast evolving digital technologies like QR code, deep-link, NFC, BLE and sound.

Use of Mobile, Instead of Plastic Card as the Channel

The Visa network is channel agnostic, and new channels like ubiquitous mobile devices add transaction capability to the network and its participants.

Visa's "product" is its electronic payment network - the transactions and the business construct that enable transfer of electronic value between stores of value (accounts) of network participants. While the predominant channel to access the network and the store of value is a plastic card, it is not the "product". These transactions are entirely mobile device led: mobile in place of plastic card and mobile (or other notification means) in place of a POS machine. The solution co-exists with the card channel for both issuers and acquirers, as depicted in the figure below, functioning on both feature and smart phones.

Mobile Device Access Co-Exists with the Card Channel



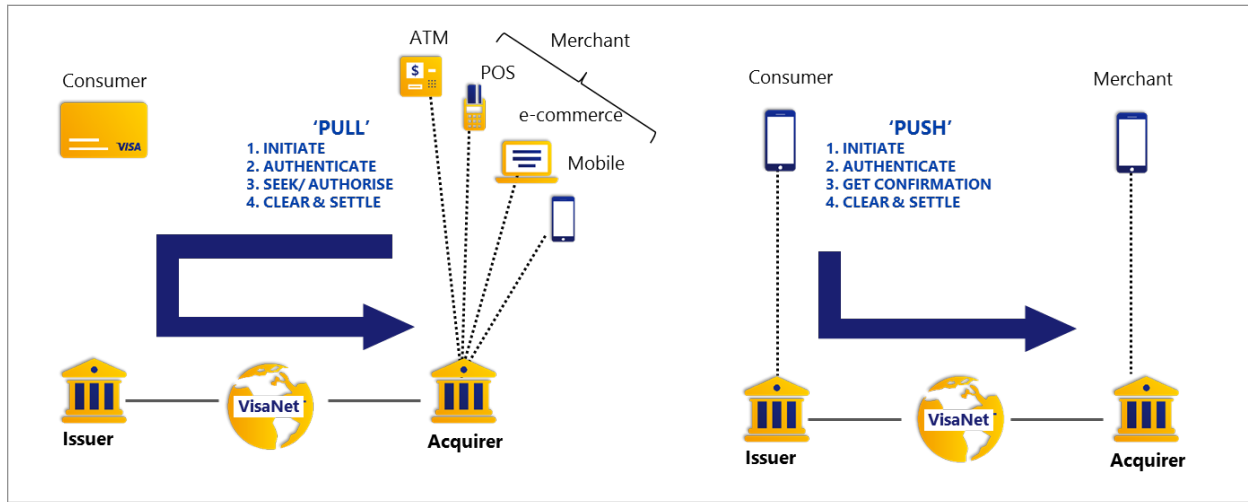
Push Payment as a Primary Transaction

To leverage the capabilities of the mobile channel better, the primary exchange of value in the solution takes place as a “push transaction”, i.e., the transaction is initiated by the **payer** first connecting to the store of value and passing instructions to transfer value to the destination of the transaction (**payee**) over the Visa network.

Hence, transactions in the solution are fully authorized and fully authenticated before initiation.

A “push” based transaction has many differences from a “pull” based transaction, and these drive the unique value proposition of the solution. See the figure and table below for more details on this difference.

Pull and Push Transactions Compared



Comparison of 'Pull' and 'Push' Transactions

Transaction component	Pull payment	Push payment
Initiation and Authorization (for funds availability/ credit worthiness)	- Initiation by the merchant. - Authorization is performed by merchant and their acquirer asking an issuer for permission to debit a cardholder's account. - Authorization may occur in different ways for different risk profiles e.g., by the issuer, by Visa stand-in or offline authorization.	- Initiated by the consumer. - Authorized by the consumer (the accountholder) by connecting directly to their account (not via the merchant). - Always authorized to ensure good funds or positive credit.
Authentication (of identity – i.e., validating that the transaction initiator is the true accountholder)	- Multiple options, each with different risk implications: - Card present Environment: online encrypted PIN, offline encrypted PIN, offline clear text PIN, signature, no CVM - Card not present Environment: Only card validation, Card and cardholder authentication using 3D Secure	Risk based authentication, with a strong method (user passcode or biometrics) by issuer.
Clearing (for accounting purpose) Settlement (actual fund paid/ received)	Clearing and Settlement are separate from authorization, so participants in the network (consumers, merchants) may experience a delay before transactions are posted to account and funds are accessible by them.	- Clearing information is provided through the SMS OCT and funds can be made accessible immediately. - Settlement continues as per local cycle.

Transactions Defined in the Program

The Mobile Push Payment program defines three interoperable push payment transactions, and includes existing person-to-person transfers, as listed in the table below.

These transactions are designed to deliver different use cases for mobile device-based access of an underlying account (store of electronic value) and exchange of value with such accounts.

Four OCT transaction types

Transaction	Transaction origination	Account identification	Business Application Identifier (BAI)
1. Merchant payment	SMS OCT, consumer mobile	Merchant ID	MP
2. Person-to-person money transfer	SMS OCT, consumer mobile	Consumer PAN	PP
3. Cash-out at merchant	SMS OCT, consumer mobile	Merchant ID	CO
4. Cash-in at merchant	SMS OCT, merchant mobile	Consumer PAN	CI

The issuer or wallet provider should preferably implement both the merchant payment and the person-to-person money transfer transactions in their program. Merchant payment transaction can enable a range of payment use cases - both face-to-face and remote.

If the originating entities do not hold the funds of the consumer, an AFT transaction with "MP" BAI can be utilized for funding the Mobile Push Payment transactions. Refer to the *Account Funding Transaction (AFT) Processing Guide* for information on how to implement AFTs. Person-to-person money transfer implies the ability to both send and receive real time (Fast funds) as defined in the *Visa Direct Original Credit Transaction (OCT) Global Implementation guide* document. Issuers and acquirers can decide to provide the other two transactions - cash-in and cash-out at merchant location - based on objectives of their mobile banking proposition.

All four transactions in the program are processed as Original Credit Transactions (SMS OCT) messages on VisaNet. However, the messages are different in how they are originated and in the account identification used for transaction routing. Specific field descriptors in each message type enable the different features required from that transaction.

All four transactions originate as full-financial, SMS transactions (i.e., "real time" messages, carrying both clearing and settlement information). Clearing information allows participants to credit (or debit) funds to involved accounts instantly. For a merchant payment, the settlement with the merchant (and therefore access to funds) can be delayed by the merchant acquirer.

VisaNet settlement between participants in the country takes place as per the settlement cycle. The four transactions are described further in the sections that follow and their features are summarized in the table "Features of the transactions in the program- a summary" in the section "Implementing Mobile Push Payment" in this chapter. The transaction steps are further described in the appendix "Transactions Descriptions" along with flow diagrams. The technical specification of each transaction along with field descriptors is provided in the *Mobile Push Payment - Technical Specifications* document.

Merchant Payment Transaction

In the merchant payment transaction, the consumer initiates the push payment to the merchant using their mobile interface (app or USSD) after information about the merchant has been “captured” on it.

The consumer is notified about the transaction status and provided a digital receipt on their mobile. The merchant is similarly notified through the notification channel enabled for them by their acquirer. This section describes the transaction as experienced by the consumer and the merchant, and the corresponding flow of the transaction as it gets processed by the issuer, over the network and by the acquirer.

Merchant Payment - Transaction Experience

The merchant payment is conducted in four steps.

These steps are explained in the figure "Merchant Payment - Transaction Experience" in the section "Capture of Merchant Information and Authentication" in this chapter. Depending on the use case, the sequence of steps could differ, or additional steps may be required.

Step 1: First, merchant information is provided by the merchant to the consumer. The provision of this information, termed as the “**Merchant information display**”, can be in various ways and of various types depending on the payment use case. The merchant information display is made available to the merchant by their acquirer and is created using specifications provided by Visa.

Step 2: Next, in the “**merchant information capture**” step, the information in the merchant display and specifically the merchant details, is “**captured**” by the consumer using their mobile phone.

Step 3: Next, in the “**authentication**” step, the consumer authenticates the transaction initiation from their underlying account using the authentication method provided by their account or digital wallet provider. Issuer can opt for risk-based authentication for their consumer. At this stage, the consumer may select their account or card. They may also input the amount to be paid if the merchant information display does not provide this information.

Step 4: Finally, in the “**notification**” step, after the transaction has been successfully processed over the network, a confirmation of payment completion is sent. The **consumer digital receipt** is received on the consumer’s mobile phone, whereas the **merchant notification** can be received on the device as enabled by acquirer.

Capture of Merchant Information and Authentication

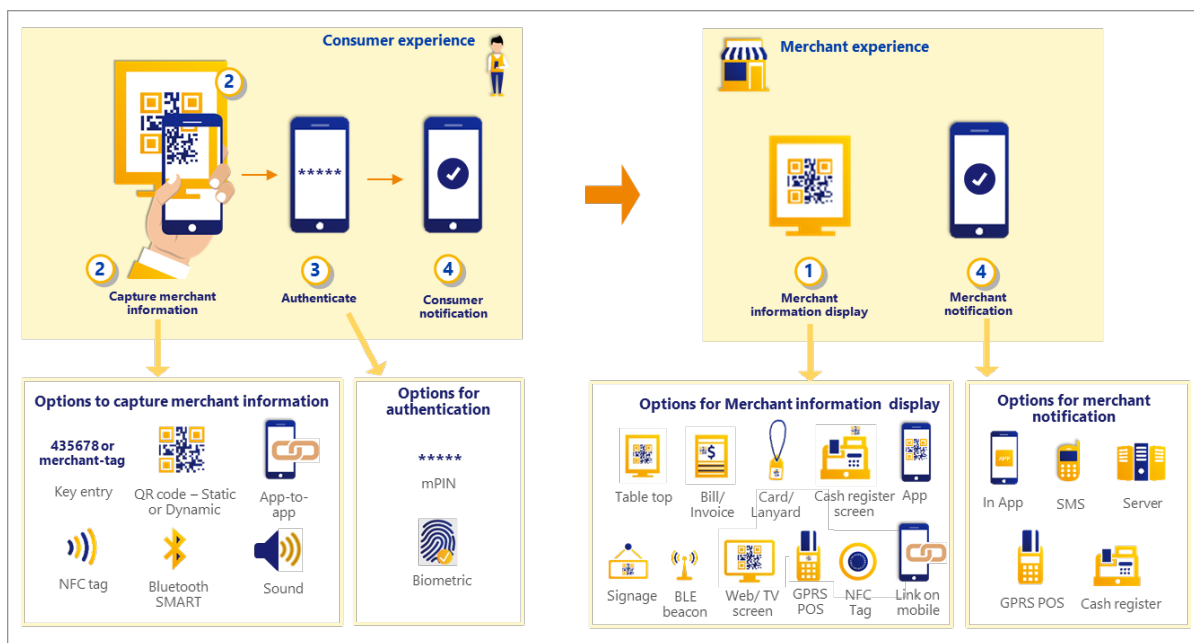
The payment instructions provided by the consumer to their issuer or wallet provider identifies the merchant PAN (which is used to route the transaction to the merchant), the amount to be paid and optionally other data about the transaction (like payment reference e.g., invoice or bill number).

This information is captured by the consumer in the payment process using the mobile interface.

As described below and depicted in figure below, the merchant information capture and the authentication method can be of various types based on the capability delivered by the issuer or wallet provider. The objective of the method enabled is to simplify the transaction experience. Details are covered in the chapter "Issuer Processes and Processing Systems".

- **Options for Merchant information capture:** This step can be conducted through the key entry of the **Merchant ID** displayed by the merchant, on the consumer mobile. This is the only method possible in the case of a feature phone. For smart phones with a mobile app, this step can utilize the camera on the mobile to scan the QR code. Alternately, information capture can utilize technologies like deep linking (app-to-app link on the same mobile), NFC, BLE or sound if made available by the merchant and on the consumers' mobile app.
- **Options for authentication:** This step can involve key entry of a numeric PIN by the consumer. The issuer could enable biometric authentication like fingerprint or other means like facial scan as such technologies become available.

Merchant Payment - Transaction Experience



Merchant Information Display and Notification

The merchant information display is enabled by the acquirer for the merchant, based on the use case and acceptance channel desired. The objective here is to simplify the transaction experience and enable new payment use cases not possible in the past. As described below and depicted in the figure "Merchant Payment - Transaction Experience", the Merchant Information Display and Notification can be of various types. Details are covered in the chapter "Acquirer Processes and Processing System". Various use cases are explained in the *Use Cases Description Guide* document.

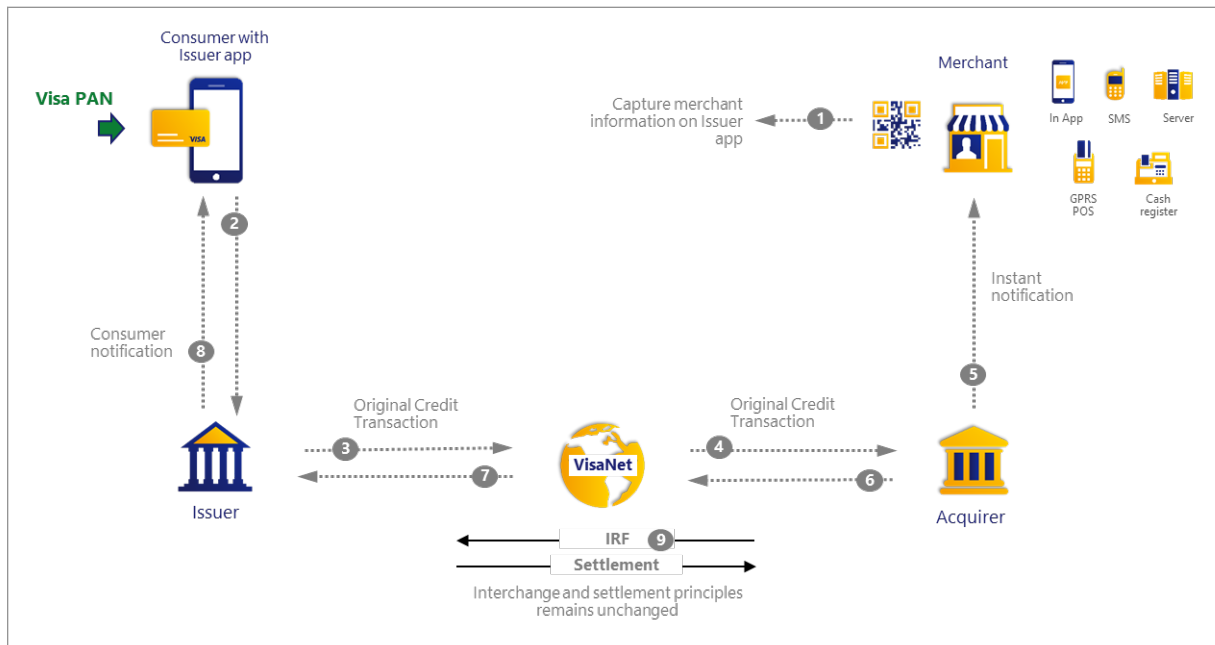
- **Options for Merchant information display:** The display can be in various ways and be of various types depending on the payment use case. It can be a static or dynamic QR code along with the Merchant ID. The QR code can be on a tabletop, printed card on a lanyard, printed on an invoice or displayed on a digital display. Other technologies like deep link app-to-app trigger, BLE beacon or NFC tag could also be deployed. The data specifications for the merchant information delivered in the merchant information display - QR code, or through any other medium like NFC, Bluetooth, or URI link - is provided in the *Mobile Push Payment - Technical Specifications* document.
- **Options for Merchant notification:** The merchant can receive the payment notification from the acquirer on their mobile or via any other channel. This channel can be the merchant till (in a multiline, integrated POS system), on a POS machine (IP connected device) or a back-end server (biller system). More than one notification channels may be deployed by the acquirer for the merchant. The various combinations possible between the Merchant ID and the merchant notification device are covered in section "Merchant Account Set-Up" in the chapter "Acquirer Processes and Processing Systems".

Merchant Payment - Transaction Flow

The merchant payment SMS OCT (0200) is initiated by the issuer to VisaNet after a consumer connects to their account through the mobile device and provides authentication and payment instructions to a merchant.

When providing the payment instruction, the consumer "captures" the merchant details from the **"Merchant Information Display"**. VisaNet routes the message to the merchant acquirer. The Visa message to the acquirer can be modified by VisaNet to suit the type of connectivity the acquirer has with VisaNet. The acquirer processes the Visa message, and if successful provides a **"Merchant Notification"**. The notification can be to the merchant mobile device or to any other notification medium that has been enabled for the merchant. The acquirer creates a record of merchant payment and may provide immediate access to their funds for the merchant. The acquirer also reverts with a response message to the issuer over VisaNet. The issuer processes the response message and triggers the **consumer digital receipt** to the consumer mobile. VisaNet settles between the issuer and the acquirer as per the settlement cycle.

Merchant Payment Transaction



Merchant PAN and Merchant ID

The Merchant PAN is created by the acquirer using a Visa assigned Issuing (ISO) BIN, acquirer assigned merchant number and check digit.

Visa uses this PAN number for routing the transaction to the acquirer processing system. Typically, an acquirer creates these Merchant PAN at the time of merchant onboarding and assigns these to a merchant or a merchant location as described in the section "Issuing (ISO) BIN Requirement and PAN Creation" in the chapter "Acquirer Processes and Processing System".

- **Visa assigned Issuing (ISO) BIN:** Acquirer is provided with a new dedicated Visa-licensed Issuing (ISO) BIN that is used to create PAN for merchants.

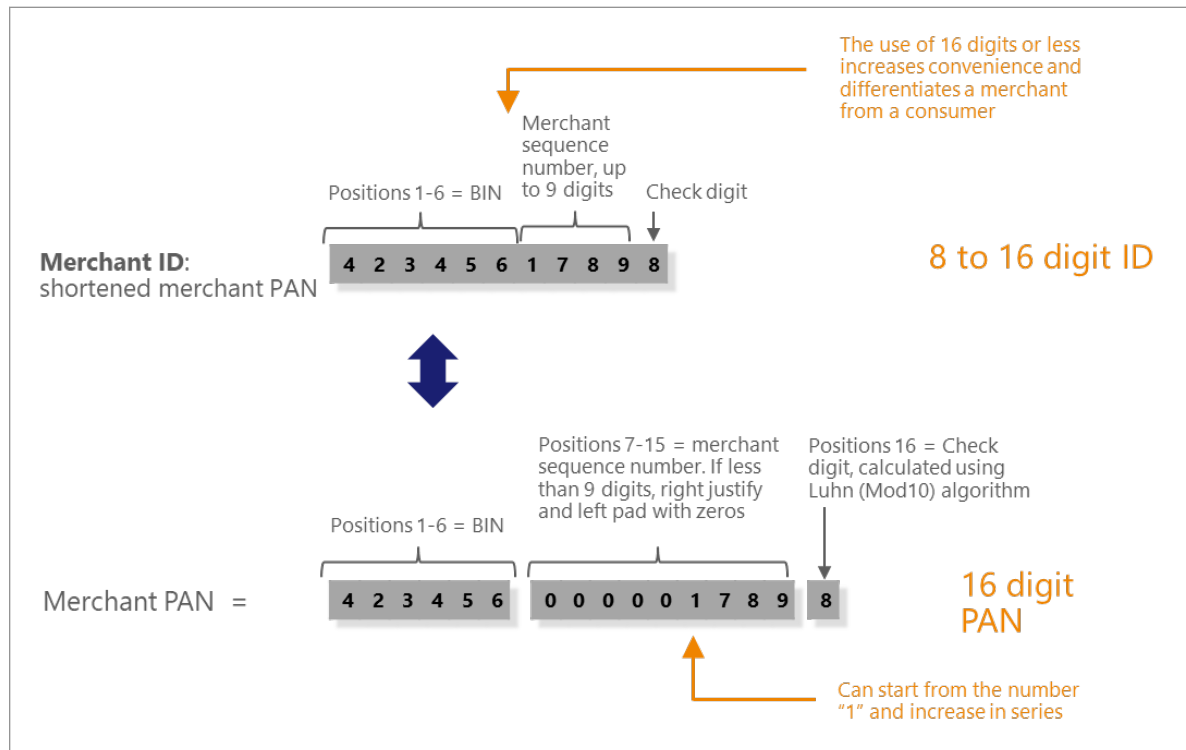
Note: Mobile Push Payment is not included in scope for the 8 digit BIN due to the construct of Mobile Push Payment usage. The merchant account is assigned under an issuing BIN but the account is not a card/credential.

- **Acquirer assigned merchant number:** Acquirer, based on its system capability and operational requirement, assigns a unique numeric value to merchant or merchant location in acquiring processing system. The acquirer may assign this in a sequential number (1, 2, 3... onwards) or any other system derived value, such that it is unique to a merchant or merchant location.
- **Check digit:** This digit is calculated using the Luhn algorithm and placed at the last digit of merchant PAN (16th digit). Luhn algorithm is also known as "modulus 10" or "mod 10" algorithm. This algorithm is the same as the one used for generating card number for Visa cards.

In some use cases, the Consumer enters a **Merchant ID** displayed on the merchant information panel. This Merchant ID is a shorter and more convenient value given by the acquirer to the merchant to provide more convenience and ease while key entering (typing), information on mobile phone. Visa supports two different type of **Merchant ID's** (a) shortened PAN and (b) alias as described below. Check with your Visa representative on applicability of these in your market.

- Merchant alias: In certain markets, Visa offer Alias Directory Services. In these markets, the acquirer assigns the merchant with alias, such as Till ID or merchant tag. These aliases are displayed on the Merchant Information Panel. The merchant PAN can be derived from these aliases.
 - Shortened PAN: In some of the early implementation, Visa provided the option of shortened PAN. Visa does not recommend the usage of these PAN's by acquirer anymore, however the Issuer app should support this for backward compatibility. The merchant PAN may also be shortened to provide convenience to Consumer. The shortened PAN is an 8 to 16-digit long number derived from the merchant PAN by their acquirer. The algorithm to create this shortened PAN from the 16-digit merchant PAN is described in the following figure. This was previously referred to as **mVisa ID**.
- I
- Issuing (ISO) BIN: Dedicated Visa-licensed Issuing (ISO) BIN or BIN range.
 - Merchant sequence number: The acquirer assigns a sequential number starting with 1 and then increments with every new merchant (1, 2, 3... onwards). It is always right-justified value while generating merchant PAN. As the merchant sequence number is less than nine digits, the vacant positions in the PAN is filled with zeros to make a 16-digit PAN. However, for display purpose these zeros are not visible in Merchant Information Panel. The issuer processing system should assume the merchant sequence number is less than nine digits and left pad the remaining positions with zeros at the time of transaction processing.
 - Check digit: As per the Luhn algorithm.

Shortened PAN for display and the Merchant PAN



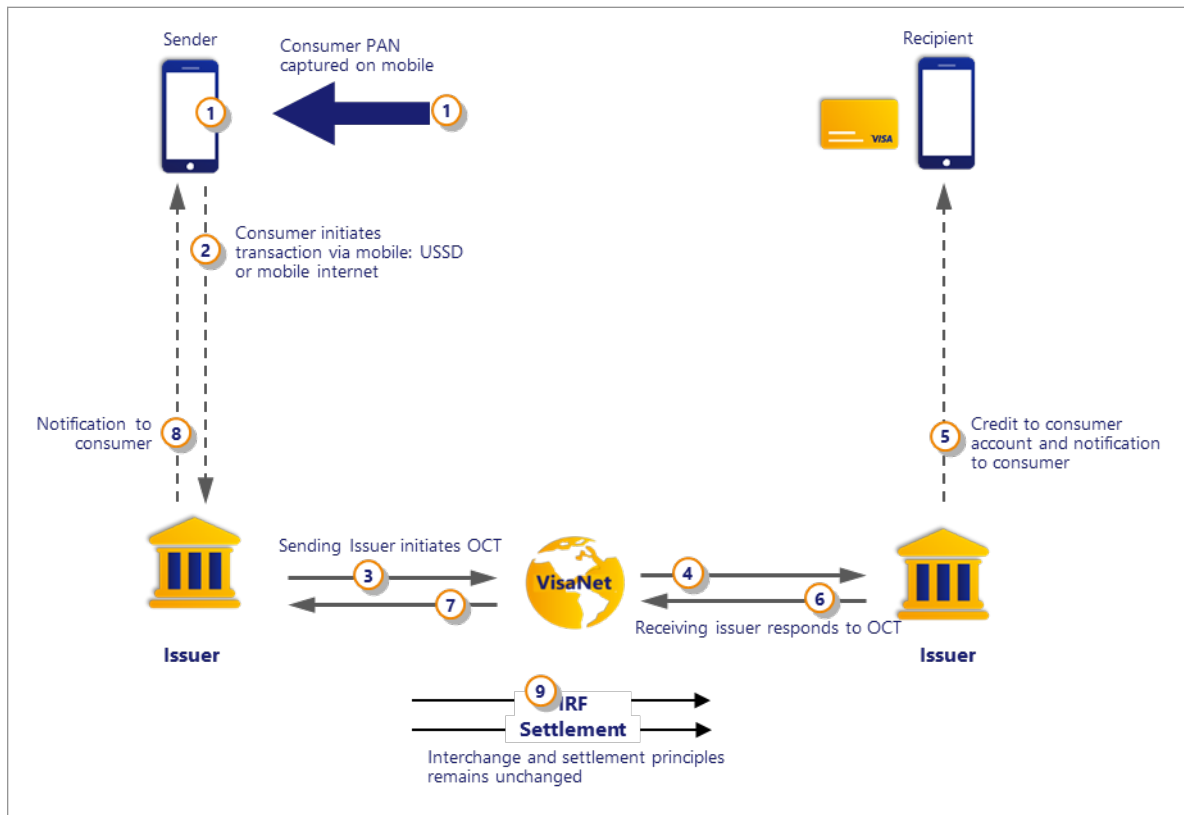
Person-to-Person Money Transfer Transaction

In this transaction the SMS OCT (0200) is initiated by the issuer when a consumer connects to their account through the mobile device, authenticates themselves and provides instructions for money transfer to another consumer.

Clients should refer to the *Visa Direct Original Credit Transaction (OCT) - Global Implementation Guide* document for implementing this transaction. VisaNet routes the transaction to the recipient issuer. The recipient issuer processes the Visa message, the receiving consumer account is credited, and a notification is provided to the recipient consumer's mobile device. The recipient issuer also reverts with a response message to the originating issuer over VisaNet. The originating issuer processes the response message and triggers a notification (confirmation or denial) to the originating consumer's mobile. VisaNet settles between the issuer and the acquirer as per the settlement cycle.

The transfer instructions provided by the originating consumer includes the recipient consumer's PAN used to route the transaction, and the amount to be transferred. This 16-digit PAN can be the same as the PAN on the Visa card that has been issued to that consumer. The structure of this 16-digit PAN is no different from the usual 16-digit PAN used to route Visa transactions.

Person-to-Person Money Transfer Transaction



Cash-Out (Withdrawal) Transaction at a Merchant

A merchant acquirer may enable cash-out (withdrawal) transaction at a merchant.

In terms of functionality, this transaction is similar to cash at POS service, but the transaction is processed differently. In the cash-out transaction, an SMS OCT (0200) is initiated by the issuer when a consumer connects to their account through the mobile device, authenticates themselves and provides withdrawal instructions. VisaNet then routes the transaction to the acquirer. The acquirer processes the Visa message, credits the merchant account if valid and reverts with a response message (0210) to the issuer. On completion, both the consumer and merchant receive confirmation or denial notifications on their mobile device. The merchant then provides cash to the consumer. VisaNet later settles between the issuer and the acquirer. The step-by-step transaction flow is described in the appendix "Transaction Descriptions", section "Cash-Out (Withdrawal)".

Cash-In (Deposit) Transaction at a Merchant

An acquirer may enable cash-in (deposit) transaction at a merchant.

In the cash-in transaction, an SMS OCT (0200) is initiated by the acquirer when a merchant connects to their account through the mobile device, provides transfer instructions and authenticates themselves. VisaNet then routes the transaction to the consumer's issuer. The issuer processes the Visa message, instantly credits the consumer account if valid and reverts with a response message (0210) to the acquirer. On completion, both the consumer and merchant receive confirmation or denial notifications. The consumer then provides the cash to the merchant. VisaNet later settles between the issuer and the acquirer. The step-by-step transaction flow is described in the appendix "Transaction Descriptions", section "[Cash-In \(Deposit\)](#)".

The transfer instructions provided by the merchant includes the depositing consumer's PAN used to route the transaction, and the amount to be deposited. This is a 16-digit PAN associated with the account by the issuer when a consumer subscribes to the service. This PAN can be the same as the PAN on the Visa card that has been issued to that consumer. The structure of this 16-digit PAN is no different from the usual 16-digit PAN used to route Visa transactions.

Branding Standards

The solution is a new channel to access the Visa network for existing or new accounts. Thus, the acceptance mark to be used on the mobile app and to be displayed at the merchant location continues to be the Visa mark.

- **Branding on the mobile interface:** Refer to the *Mobile Push Payment -Mobile User Experience Guide (Mobile App, USSD)* document for branding elements that have to be included by clients and partners in the transaction experience on the mobile. The document provides brand requirements on the mobile interface, brand requirements in product communication and brand specifications.
- **Branding at the Merchant "location":** Refer to the *Mobile Push Payment Merchant Information Display Guide* document for the branding elements that have to be included by acquirers and partners to facilitate transaction experience at merchants. The document provides brand requirements on the merchant information panel, brand requirements in product communication and brand specifications.

Program participants are required to submit branded elements like the mobile app screens and merchant information panel to Visa for approval, before deploying them.

Mobile User Experience Guidelines and Mobile SDKs

The success of the client's program is critically dependent on a good and consistent mobile user experience.

Therefore, program participants should follow the guidelines provided by Visa for design of the mobile interface and transaction steps on the mobile app.

- | These guidelines for issuers, acquirers and partners are provided in the *Mobile Push Payment - Mobile User Experience Guide* document. They include the end-to-end mobile screen flows with description. To help deploy the experience, issuers and acquirers should use the mobile SDK provided by Visa and described in the *Mobile SDK Integration Guide* document.

Program participants are required to submit the mobile user experience they intend to deploy to Visa for approval before its implementation and market deployment.

Settlement

Domestic transactions in the program are settled in local currency, using VisaNet's National Net Settlement Service (NNSS) already in use in the country.

International transactions in the program are settled using VisaNet's International Settlement Service. Settlement is conducted once daily or more frequently based on the set-up of NNSS. A summary of the settlement applicable for the transactions in the program is listed in table in the section "Interchange Reimbursement Fees (IRF)" in this chapter.

Interchange Reimbursement Fees (IRF)

- | There is no change in how IRF is applied by VisaNet to transactions in the program: acquirer pays issuer in the merchant payment transaction; originating client pays receiving client in the person-to-person money transfer transaction; issuer pays acquirer for cash transactions.

The cash-in and cash-out transactions at the merchants are treated like transactions at an ATM – with the issuer paying the acquirer in both transactions. A summary of the IRF applicable to various transactions is listed in the following table. Clients should contact their Visa Representative to obtain the IRF rates applicable for their market.

Summary of Settlement and IRF Flows

Transaction	Settlement	Interchange (IRF)
1. Merchant payment	Issuer -> Acquirer	Acquirer -> Issuer
2. Person-to-person money transfer	Sender's provider -> Recipient's provider	Sender's provider -> Recipient's provider
3. Cash-out at merchant	Issuer -> Acquirer	Issuer -> Acquirer
4. Cash-in at merchant	Acquirer -> Issuer	Issuer -> Acquirer

VisaNet Pricing

Clients should contact their Visa Representative to obtain the VisaNet transaction pricing applicable for their market.

Quarterly Operating Certificate

The program does not represent a new product and is only a new channel to access the Visa network.

Hence, there is no change in the QOC reporting format or requirements for the program.

- **For issuer:** The issuer can enable their consumer accounts with physical or virtual Visa credentials. This can be a debit, credit or pre-paid card or credential. The transactions processed have to be reported and included as per the existing QOC format - i.e., if the issuer has assigned the consumer a debit platinum card or credential, then the merchant payment volume is reported under debit platinum sales volume.
- **For acquirer:** The acquirer has to report and include the transaction processed as merchant payment volume in the usual QOC format.
- **Cash-in and cash-out:** The cash-in and cash-out transactions have to be reported and included under cash volume on respective cards by the issuer and by the acquirer.

International Transactions

- I The program supports cross-border merchant payment, person-to-person money transfer and cash-out.

Cross border cash-in transactions are not allowed in the program. Such transactions are settled internationally by VisaNet. The requirements for enabling international transaction are listed below:

1. The consumer originating the transaction is always debited in his or her own account currency.
2. The consumer initiating the transaction has to be made aware of, and should accept, their participation in the international transaction.
3. The originating issuer determines the recipient account currency based on the recipient PAN currency. There are several ways that an originator can obtain details about the recipient currency.
 - a. **QR Code:** QR code contains the transaction currency of the merchant; during the scanning process, the app can determine the currency of the merchant.
 - b. **Account Lookup API:** Visa provides account lookup API. These API returns to the origination information such as the country code and billing currency, account type etc. for the merchant PAN. Information on this API is available at developer.visa.com.
 - c. **Account Lookup File:** If the client does not prefer API, same information on the recipient PAN is available in a file format - '**TC 54 ARDEF file**'. This file is available using API (details on developer.visa.com) or through Visa File Exchange Service (VFES).
4. Once the recipient currency is known, the originator has to determine the forex rate to apply between the consumer's currency and merchant's currency. The issuer can use their own rate or refer to Visa provided rates available through TC 33 or API query.

All originators are required to support cross-border merchant payment. Refer to the section "Merchant Payment Transaction" in the appendix "[Transaction Descriptions](#)" for step-by-step process for international cross-border merchant payment. Originators may contact their regional Visa representative to obtain more details on the information sources.

Dispute Resolution

The possibility of disputes in the program is significantly reduced due to the nature of the transactions in the program.

The table below summarizes the dispute scenarios applicable for transactions in the program.

- “Push transactions” are payer initiated and authenticated at origination by the user first connecting to their account. Thus, disputes due to fraud (stolen account) or “I did not authorize or participate in the transaction” are not an issue to be solved through Visa’s Dispute Resolution process.
- Any uncertainty of receipt of funds at the receiving client (merchant or consumer account) is removed through the mandatory response message from the receiving system (0210 message).

Summary of Dispute Management Scenarios Available

Transaction	Issuer	Acquirer
1. Merchant payment.	Originating consumer’s account provider can dispute a transaction for the Dispute Conditions as described in the section “Risk mitigation by all Program Participants” in this chapter.	Merchant’s account provider: Respond to dispute as stipulated in the <i>Visa Core Rules</i> and <i>Visa Product and Service Rules</i>
2. Cash-in at merchant.	Recipient issuer, no reversal and disputes rights applicable.	Originating acquirer - no dispute rights applicable
3. Cash-out at merchant.	Originating issuer, no dispute rights applicable.	Recipient acquirer, no reversal or dispute rights applicable.

Dispute Conditions for Merchant Payment Transaction

- The Dispute Conditions available to the issuer for the merchant payment transaction in the program are explained in table below.

Issuers and acquirers must refer to the *Visa Core Rules* and *Visa Product and Service Rules* for additional requirements on how to support or remedy the applicable dispute condition.

Note: Any domestic rules noted in these dispute conditions may not apply for disputes.

Applicable Dispute Conditions for Merchant Payment Transactions

#	Dispute Condition
1	12.6 (Duplicate Processing/Paid by Other Means)
2	13.1 (Merchandise/Services Not Received)

Applicable Dispute Conditions for Merchant Payment Transactions

#	Dispute Condition
3	13.3 (Not as Described or Defective Merchandise/Services)
4	13.4 (Counterfeit Merchandise)
5	13.5 (Misrepresentation)
6	13.7 (Cancelled Merchandise/Services)

The following additional considerations are applicable for merchant payment transaction:

#	Scenario	Additional Consideration
1	In case of key entered transaction, Consumer enters incorrect merchant information and transaction is completed	Issuer cannot dispute such transaction, the only recourse available for Consumer and Issuer is to follow the good-faith dispute with merchant and acquirer.
2	Merchant does not receive notification, Consumer receives successful digital receipt	Merchants are required to resolve the dispute by checking their last few transactions or contacting the acquiring bank to ascertain the status of the transaction. Dispute Condition 13.1 - Merchandise/ Services Not Received may apply; however, the Issuer is required to collect a signed consumer letter in case the consumer submits more than three disputes in 30 days period to act as a deterrent for misuse.
3	Incorrect amount credit to merchant, under payment or excess payment	If the merchant is under paid, the consumer can initiate a transaction for the balance amount. In case merchant is over paid, then the consumer can request merchant to refund by directly crediting the difference into consumer PAN or by other means.

Dispute Resolution Process - General Requirement

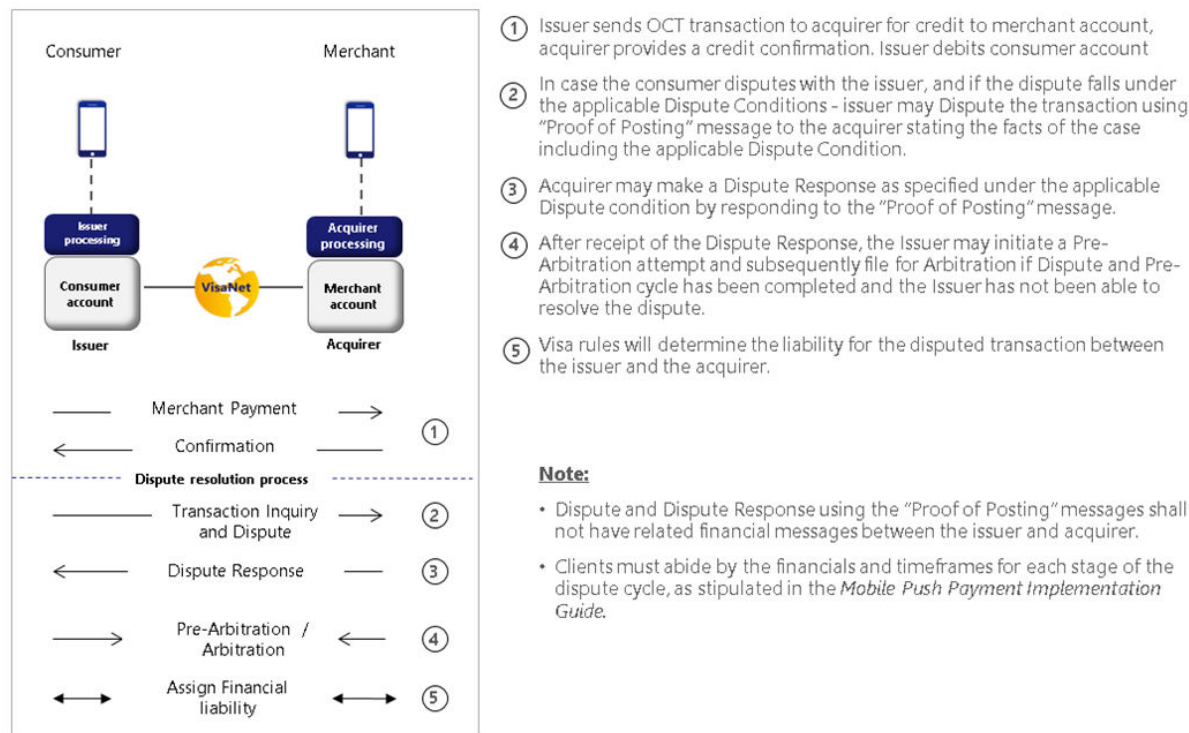
The Visa Resolve Online (VROL) system from Visa supports the dispute resolution process for a Mobile Push Payment as outlined below.

Clients must refer to Chapter 11 of the *Visa Core Rules and Visa Product and Services Rules* for the Dispute Resolution Rules and requirements. The resolution process for a merchant payment transaction as depicted in the figure below.

Dispute Resolution Process

Dispute resolution for Merchant Payment Transaction

Dispute resolution process as administered via VROL



Dispute Resolution Process

A Member must comply with the following table for processing Mobile Push Payment Disputes and related activity.

Refer to Chapter 11 of the *Visa Core Rules and Visa Product and Services Rules* for the Dispute Resolution Rules and requirements.

Dispute Resolution Process

Dispute Process Stage	Description
Dispute Time limit: See applicable Dispute condition	After receiving the complaint from consumer, an Issuer may initiate a Dispute under the conditions specified for the applicable Dispute category. The Issuer must detail the Dispute Condition in the Proof of Posting Message as follows: Dispute Condition XX1 – Description ***** <free text for originator to provide> *****
Dispute Response Time limit: 30 calendar days from the Dispute Processing Date	In response to a Dispute, the Acquirer may make a Dispute Response as specified under the applicable Dispute condition.
Pre-Arbitration Attempt Time limit: 30 calendar days from the Dispute Response Processing Date	After receipt of a Dispute Response, the Issuer may make a pre-Arbitration attempt for any of the following reasons: The Issuer can provide new documentation or information to the Acquirer about the Dispute. If the Acquirer has supplied Compelling Evidence, the Issuer certifies that it has contacted the Cardholder to review the Compelling Evidence and either: In the Europe Region, can provide information detailing how the Compelling Evidence has been addressed by the Cardholder and why the Cardholder continues to dispute the Transaction For all other Transactions, can provide an explanation of why the Cardholder continues to dispute the Transaction. The Issuer changes the Dispute condition after receiving the Dispute Response. If the Acquirer provided evidence that the Cardholder no longer disputes the Transaction, the Issuer certifies that the Cardholder still disputes the Transaction.
Pre-Arbitration Response Time limit: 30 calendar days from the Processing Date of the pre-Arbitration Attempt	An Acquirer may respond to the pre-Arbitration attempt as follows: The Acquirer may accept financial responsibility. The Acquirer may decline the pre-Arbitration attempt. If the Acquirer accepts financial responsibility, they must credit the Issuer for the last amount received by the Issuer within 30 calendar days from the Processing Date of the pre-Arbitration Attempt
Arbitration Time limit: 10 calendar days from the Processing Date of the pre-Arbitration response	The Issuer may file for Arbitration when one of the following occurs: The Dispute and Pre-Arbitration cycle has been completed and the Issuer has not been able to resolve the dispute. The Acquirer has not met the requirements specified in the Visa Rules.

1 Where XX.X is the Dispute Condition number

Arbitration or Compliance Financial Liability

This section shows results for an Arbitration or Compliance.

An Arbitration or Compliance decision may result in either:

- One Member assigned full liability
- Members sharing financial liability

The responsible Member is financially liable for the following:

- Transaction amount. If the opposing Member is responsible, the requesting Member may collect this amount from the opposing Member within 60 calendar days from the Notification date of the decision by the Arbitration and Compliance Committee.
- Review fee

When the case is adjudicated, Visa will collect the review fees through the Global Member Billing Solution from the responsible Member.

Risk Management

Program participants should actively manage and mitigate risk in the network by adhering to the safeguards outlined in this section.

- | For additional information on risk management for person-to-person transactions, refer to the *Visa Direct Original Credit Transaction (OCT) – Global Implementation Guide* document.

Risk Management by all Program Participants

This section outlines various forms of risk management that are to be practiced by all program participants.

- **Apply velocity limits:** Apply transaction velocity limits in their processing system. The section "Enabling Risk management controls" in the chapter "Acquirer Process and Processing System" describes how merchant acquirers can also enable value and velocity controls at VisaNet.
- **Protect user data:** Follow and implement the Payment Card Industry Data Security Standards (PCI DSS) for protection of participant information.

- **Send digital receipt to consumer and merchant notifications:** Communicate transaction confirmation and status message to the users for every transaction, through the mobile device or any other notification method. Issuers are required to provide merchant information in the digital receipt to consumer as provided in the response message from the acquirer and not as available from the merchant information panel.
- **Transmit key data:** Include data elements or information needed to support regulatory compliance (e.g., sender data, source of funds, velocity data) in the transaction.

Risk Management by Issuers in the Program

This section outlines various forms of risk management that are to be practiced by issuers in the program.

- **Follow regulatory standards:** Ensure that the program includes adherence to know-your-customer (KYC) requirements, and appropriate safeguards for fraud, anti-money laundering (AML), anti-terrorist financing (ATF), and suspicious activity monitoring and reporting. All origination “send” programs must register and be approved by Visa before launch and satisfy applicable KYC requirements with respect to its Visa cardholders.
- **Apply limits to the transaction size:** VisaNet blocks OCT of value greater than \$50,000. In addition, issuers should limit the maximum value of a transaction in the program to comply with applicable local laws and regulations, and consistent with their own risk management policy. Issuers should also consider setting optional velocity limits on transactions based on parameters such as timeframe and geographic scope.
- **Perform user authentication:** For all transactions, clients must use a risk-based authentication mechanism (e.g. numeric PIN, password, biometric) to authenticate users and confirm that the transaction originates from a pre-registered mobile number or mobile device (device fingerprinting).
- **Decline or Dispute transactions for inappropriate accounts:** For person-to-person transactions, recipient issuers should decline transactions to accounts that have not fulfilled KYC requirements (or do not satisfy other regulatory requirements), to accounts that have been closed, and if the recipient refuses the credit (Dispute).
- For additional guidance on risk management for issuers, please also see the section [“Enabling Risk Management Control.”](#)

Risk Management by Acquirers in the Program

This section outlines various forms of risk management that are to be practiced by acquirers in the program.

- **Implement ability to track transactions:** Acquirer processing system should be capable of tracking, detecting, and preventing fraud. In case the acquirer support cash-in and cash-out transaction then the acquirers in particular should be capable of identifying suspicious transaction activity (e.g. transaction gaming).
- **Provide merchant training:** Acquirers must have training program related to cash-in and cash-out processes, if enabled at the merchant location.
- For additional guidance on risk management for acquirers see the section "[Enabling Risk Management Control](#)".

API Integration with VisaNet

To implement the program, both issuers and acquirers have the option to integrate with Visa either via ISO based connectivity or via Visa APIs.

Specification of the Visa APIs and integration manuals are available on developer.visa.com. Visa offers API's to originate (issuer side) and receive (acquirer side) and for different transactions (merchant payment, person-to-person, cash-in, cash-out). Domestic or international transactions can be enabled using APIs. The client can decide to use any one or all the available API's during the implementation.

Mobile SDKs

To accelerate the development of their mobile app, issuers and acquirers can utilize software development kit (SDKs) provided by Visa.

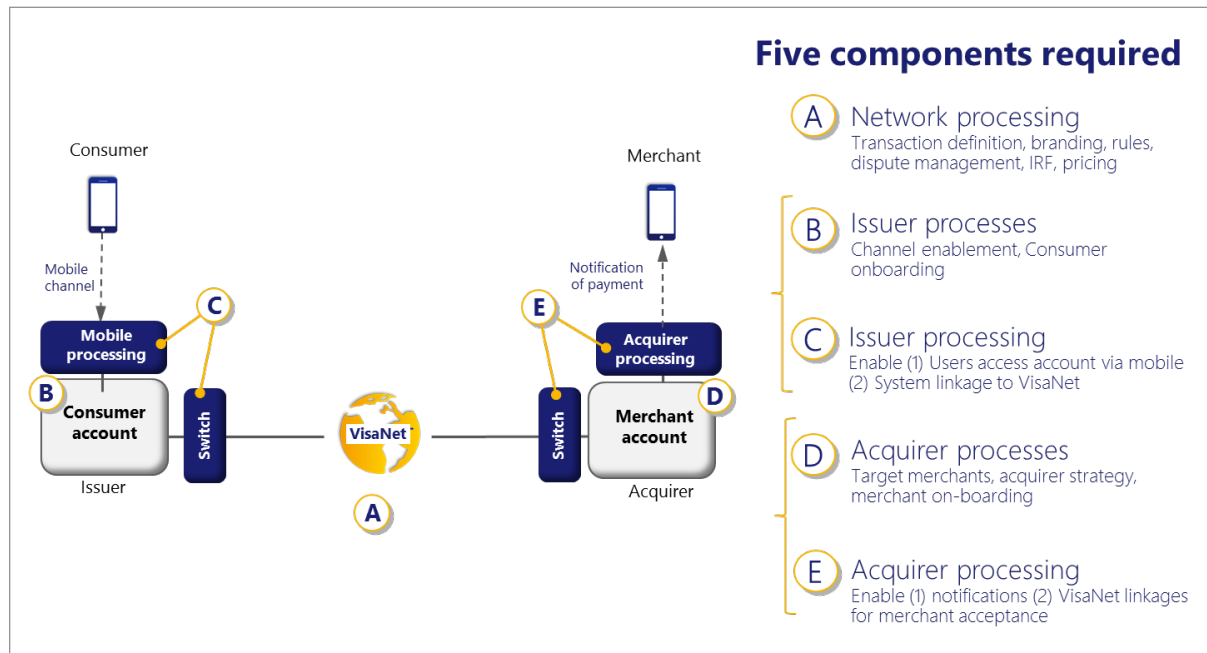
The SDKs include pre-packaged UI screens and code for the mobile transaction experience (merchant payment and person-to-person use cases). The SDKs are based on Visa's UI/UX guidelines for the transaction flow and clients have to develop their own customer onboarding and account management functions. SDK customization ability allows seamless integration between push payment specific component and existing bank app. Designed with Human Centric Design principles, the SDKs offers an intuitive UX and supports Apple (iOS) and Android on various device manufacturers. The SDK's can be downloaded from developer.visa.com.

Implementing Mobile Push Payment

To roll out their program, issuers and acquirers are required to implement operational processes and deploy processing capability based on defined network standards.

These requirements can be described under five broad headings as depicted in the figure "Enabling cardholder via Mobile App" in the section "Consumer On-Boarding" in the chapter "Issuer Processes and Processing System".

Solution Architecture and Implementation Components



Issuer processes are those that enable the mobile channel for the consumer and onboarding of the consumer. **Issuer processing** is the capability of a financial account to receive transactional instructions from a consumer's mobile device via the mobile channel and notify them, along with the ability to send and receive transactional messages from VisaNet. Issuer processes and processing system requirements are covered in the chapter "[Issuer Processes and Processing System](#)".

Acquirer processes are those that enable merchant targeting, sign-up and onboarding. Acquirers have to enable merchants with a merchant display and a notification method. **Acquirer processing** is the capability to receive and act up on transactional messages received from VisaNet and provide notification to merchants for payments received. Acquirer processes and processing are covered in the chapter "[Acquirer Processes and Processing System](#)".

Features of the transactions in the program - a summary

Transaction	Account ID/PAN	VisaNet Origination	Initiating Device	VisaNet Settlement	VisaNet IRF	IRF Value	VisaNet Pricing	Dispute Rules
Merchant payment	Merchant ID/ converted to PAN	SMS OCT (BAI = MP)	Consumer mobile	Issuer TO merchant acquirer	Merchant acquirer TO issuer	Existing	Existing pricing	• Issuer = no reversal • Issuer can dispute for applicable reasons
Person-to-person money transfer	Consumer PAN	SMS OCT (BAI = PP)	Sender's mobile	Sender TO recipient	Sender TO recipient	Existing	Existing pricing	• As described in, <i>Visa Direct Original Credit Transaction (OCT) - Global Implementation Guide</i>
Cash-out (withdrawal)	Merchant ID/ converted to PAN	SMS OCT (BAI = CO)	Consumer's mobile	Issuer TO acquirer	Issuer TO acquirer	New, in NNSS	New	• Issuer = no reversal • Acquirer = no disputes applicable
Cash-in (deposit)	Consumer PAN	SMS OCT (BAI = CI)	Merchant mobile	Acquirer TO issuer	Issuer TO acquirer	New, in NNSS	New	• Acquirer = no reversal • Issuer = no disputes applicable

Chapter 3

Issuer Processes and Processing System

This section describes the business processes and the processing system capability that are required by the issuer in the program.

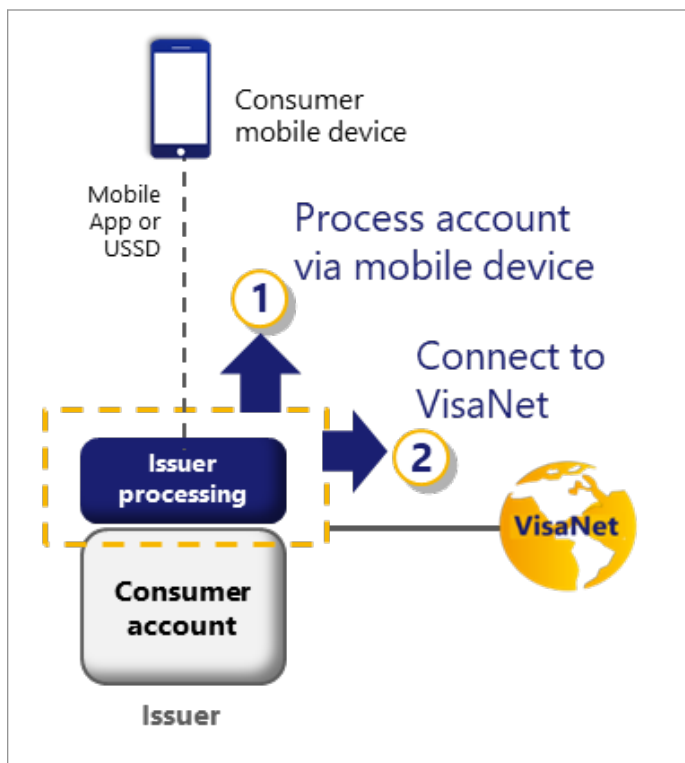
It explains functionality required to enable the mobile channel. Processes like consumer onboarding and PAN management are described. Functional components required in the issuer processing system and their possible configuration have been explained.

Functionality Required by Issuer

To participate in the program, an issuer needs processes and processing capabilities that can be grouped under two broad areas as shown in the figure below.

- 1. Processing of the account through mobile channel:** Consumer interacts with their account in the issuer system via a mobile interface like mobile app or USSD. Instructions given by the consumer or merchant data captured from the merchant information display on the mobile translate into actions in the processing system. Other processes include consumer authentication, consumer on-boarding and PAN management in the processing system.
- 2. Processing of VisaNet transactions:** Issuer processing system creates VisaNet transaction messages based on the input provided by the consumer via their mobile device. It also processes transactions received from VisaNet into actions in the processing system.

Two Broad Issuer Processing Capabilities



Processing the Account through the Mobile Channel

Issuer has to select the **mobile channel** through which consumers will access the underlying account in the program.

The most common mobile channels used for financial transactions are mobile app and USSD. The issuer is likely to be already providing services using at least one of these channels.

The issuer has to decide the **underlying account** to which they will provide functionality – savings account, current or checking account and line of credit. The underlying account can be existing account or new account. The existing account may already have a Visa card linked to it – debit card (savings account), commercial card (current account) or a credit card (line-of-credit).

Decision on Mobile Channel

The issuer's choice of the mobile channel to deploy for their program is based on multiple factors: penetration of smart phones in the issuer's market; commercial availability of USSD in the market; issuer's channel strategy; and the segments that the issuer is targeting with this solution.

The following table provides a comparison between deployment through mobile app and USSD.

Mobile App and USSD Compared

Comparison	Mobile App	USSD
Mobile device	Smart phone	Smart phone or Feature phone
Technology Requirements	• App downloaded on mobile device. Back end server to manage the app. • Consumers need to have a data connection to use. • Application needs to be compatible with the handset.	• Common USSD short code from all MNOs and integration with MNO. • Back end server to deliver the menus and manage the USSD session. • Session based – so needs unbroken connectivity during the transaction.
User interface to capture payment information	Various - Key entry, Scan QR code, Tap NFC tag (if NFC phone), Wave Bluetooth (if Bluetooth enabled), sound, app to app.	Key entry only
Data Transfer	Can be encrypted	Session secure but data unencrypted

Mobile App and USSD Compared

Comparison	Mobile App	USSD
Deployment options	- Existing issuer mobile app now with new payment option included. - Or new, dedicated app for payments.	- Existing issuer USSD menu now with the payment option included. - New, dedicated USSD short code and backend menus for payments.
User Experience	- Easy menu-based interface. - Data transfer can be encrypted. - Ease of use since application is resident on device.	- Not true menu – requires user to “answer” with their menu selection. - Works ‘out of the box’ without any downloads needed.

Mobile App

This section outlines issues related to mobile apps.

Mobile App Development

The issuer can make push payments available through a mobile app in one of two ways.

- 1. As part of issuer’s existing mobile app.** Here the existing, already published app is updated by the issuer with Visa transaction screens. The advantage of this option is that the consumer does not have to take any steps to obtain access to Mobile Push Payment.
- 2. As a dedicated new app.** Here a new app is published by the issuer dedicated for payment transactions. The consumer has to take the additional step of downloading the app. However, the transaction experience stands out and is not missed amongst many other services.

Issuers will have to take a decision regarding the number of app platforms (e.g., Android, iOS, or others) through which they will make their app available in the market.

Mobile App User Experience

When designing and creating the mobile screens or the dedicated app for Mobile Push Payment, the issuer should keep the following user experience related points in mind.

The issuer may utilize the mobile SDK provided by Visa to simplify their implementation effort.

- The user experience and **mobile screens** to be used for Mobile Push Payment transactions have been defined in the *Mobile Push Payment – Mobile User Experience Guide*. These have been created after careful evaluating of options, speed of conducting transactions and testing in consumer research.
- Issuer can modify the user experience and screens within the guidelines. Any modification in the mobile screens should be designed with the objective of simplifying the transaction experience for the user. The look, feel and colors of the screens can also be modified to comply with the issuer's brand guidelines. The final screens should be submitted to Visa for approval before being coded into an app for publishing.
- Among the mediums enabled for **capture of merchant information**, the issuer must provide "key entry" and "QR code scan" in their app. These two options are likely to be available from acquirers at all enabled merchants.
- Issuer should place particular attention to **authentication** in the payments flows since that step impacts the user experience significantly. The ease and time taken to complete the transaction will influence their adoption. User experience has to be in balance with security considerations. Since the processing system is also required to authenticate the device from which the transaction originates, a numeric PIN authentication provides **two-factor authentication**. This numeric PIN can be replaced by some other second factor like a thumbprint.
- **Digital receipts** are an important component of the payment experience. In addition to the in-app messages, issuers can consider sending digital receipt through additional options like SMS and email.
- To enhance the consumer experience, other account related information should be included in the mobile screens. **Available balance** (or available line of credit) is one such information that is important for users. Quick access to **recent transactions** is also very useful. For Visa payments, the transaction tracker should include basic information like transaction date, merchant name and amount.
- Issuers should leverage the opportunity provided by the mobile channel to deliver **value added services** to consumers via the mobile app, e.g., real-time spend data analysis, offers and loyalty alerts. The scope for innovative services is wide and now available to be tapped using the app.

USSD Channel

This section outlines issues related to the USSD channel.

Deploying USSD Channel

The issuer can make transactions available through USSD in one of two ways:

1. **As part of issuer's existing USSD short code.** Here the menu served when the existing USSD short code is dialed, is updated with the Visa payment option. The advantage of this option is that the consumer does not have to recall a new short code or register using a new short code.
2. **As a dedicated USSD short code with Visa payment menus.** Here a new USSD short code is obtained by the issuer and the USSD menus are Visa focused. The consumer has to take the additional step of registering using the new USSD short code. However, the transaction experience has fewer and more focused transaction steps.

USSD Channel User Experience

The issuer should engage with their Visa Implementation Manager when designing the USSD menus and the USSD screens.

The issuer should keep the following user experience related points in mind:

- The USSD screens to be used for the transactions have been defined in the *Mobile Push Payment – Mobile User Experience Guide*. These have been created after careful evaluating of options, speed of conducting transactions and have also been tested in consumer research.
- Issuer can modify the experience within the guidelines. Any modification in the USSD menus should be designed with the objective of simplifying the transaction experience for the user. The final screens should be submitted to Visa for approval before being published.
- Since the processing system is also required to authenticate the device from which the transaction originates, a numeric PIN authentication provides two-factor authentication.
- Digital receipts are an important component of the user experience. In addition to the sending receipts through SMS message, issuers can consider including additional options like USSD flash messages and email.
- To enhance the consumer experience, other account related information should be made available through the USSD menu. Available balance (or available line of credit) is one such information that is important for users. Quick access to recent transactions is also useful. Issuers can decide to provide only Visa push payment transactions or include other transaction types as well. The Visa transaction tracker should include basic information like transaction date, merchant name and amount.

Consumer Onboarding

This section outlines methods of Onboarding consumers.

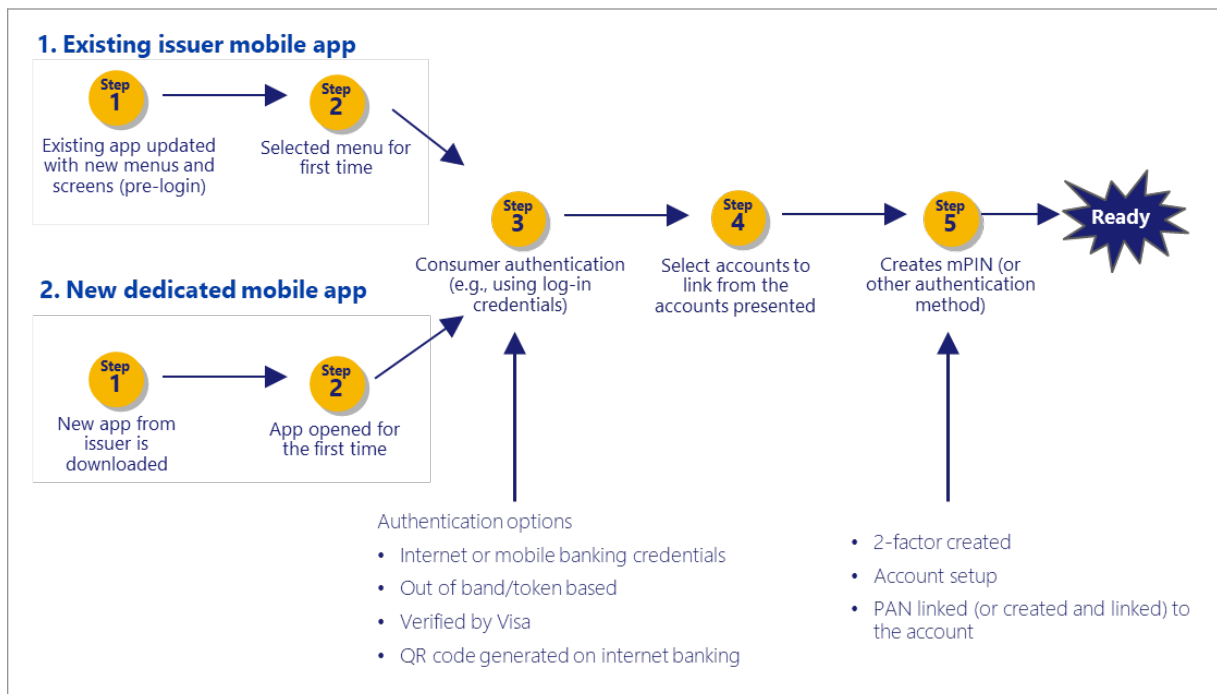
Onboarding Consumers via the Mobile App

Regardless of how the app interface is made available to consumers, to be able to conduct transactions, the consumer has to take the first step of linking their account with the app.

During this onboarding step, a Visa PAN is linked to the account to identify it uniquely. If the account has existing access via a Visa card, the Visa PAN can be used for this purpose. If not, the issuer first generates the Visa PAN (virtual card issuance) which is linked to the account.

Issuers can select various ways to onboard and enable the account linkage. To drive adoption, it is important that this step is made as easy as possible. The best route is to present all accounts that the consumer has with the issuer, so that the consumer can make their selection. For a new customer, the account will have to be first created in the system. The suggested steps in the onboarding process are outlined in the figure below.

Enabling Cardholder via Mobile App



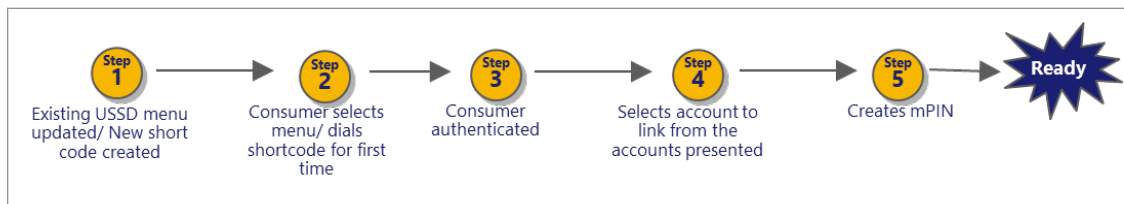
Onboarding Consumers via the USSD Channel

Regardless of how the USSD interface is made available to consumers, as a first step the consumer has to be on-boarded in which the consumer selects their account to be linked and creates a numeric PIN.

A new consumer will need to first enroll at an agent or a branch to open an account in the system. During the onboarding step a Visa PAN is linked to the account to identify it uniquely. If the account has existing access via a Visa card, the Visa PAN can be used for this purpose. If not, the issuer generates the Visa PAN (virtual card issuance) to be linked to the account.

To drive adoption, it is important that the onboarding step is made as easy as possible. The most effective route is to present all accounts that the consumer has with the issuer, so that the consumer can make their selection. The suggested steps in the onboarding process are outlined in following figure.

Enabling Cardholder via USSD Channel



PAN Management in the Processing System

The issuer processing system should be able to create, map and retrieve the PANs as required by the program.

For a consumer account that is on-boarded, the processing system maps the consumer PAN with the account. To identify the recipients of push transactions, the system populates the PAN into the VisaNet transaction from the mobile interface as captured by the consumer.

Coexistence of Card and Mobile Payment

Issuers can give access to the same underlying account using both the Visa card and Mobile Push Payment transactions.

Consumers will be able perform merchant payments with either the card or through the mobile channel - in addition to the other Visa transactions enabled by the issuer through mobile.

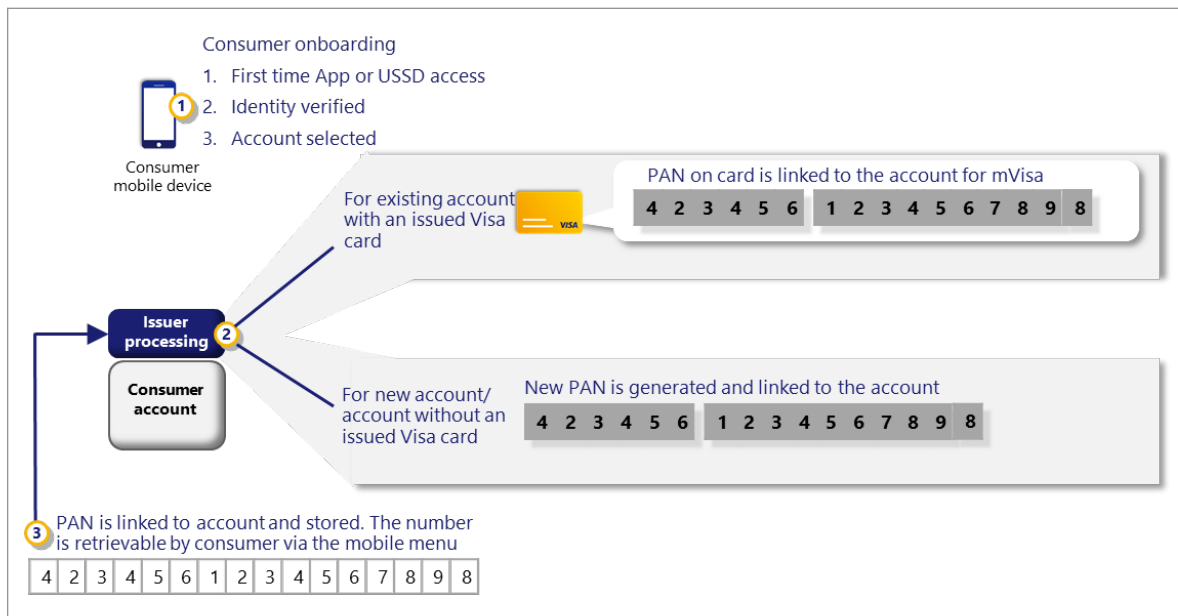
Creating and Linking of PAN with Account at Consumer Onboarding

Visa consumers are identified using 16-digit Visa PAN in the transactions.

Consumers with existing Visa plastic card can be identified using their available Visa card number. Consumers without a Visa card, may be issued a 16-digit Visa PAN (virtual card) on a new Issuing (ISO) BIN or on any of the issuer's existing Issuing (ISO) BIN that identifies the consumer segment appropriately (basic account or premium account). Issuer must use their Issuing (ISO) BIN appropriate to the underlying account.

During the consumer on-boarding process, a personal account number (PAN) is linked to the account by the issuer processing system. The step wise process is depicted in the figure below. Consumer should be able to access their PAN number) from their mobile interface after authentication.

PAN Linkage at Consumer Onboarding



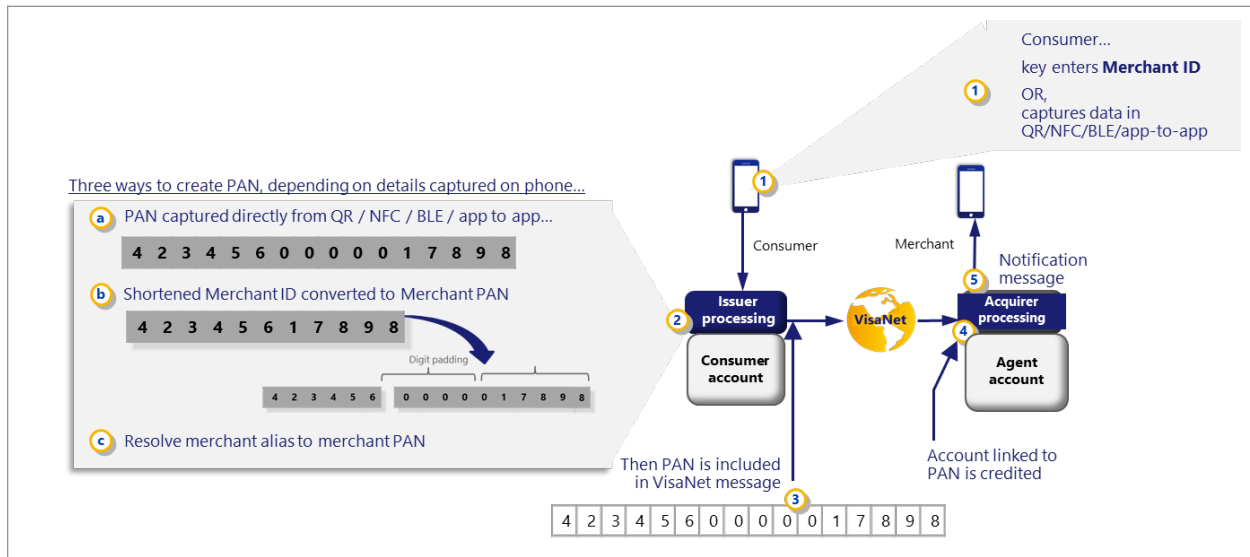
Creating a Merchant PAN from the Merchant ID

Visa consumers are identified using 16-digit Visa PAN in the transactions.

During the merchant payment or cash-out transactions, the merchant PAN has to be determined by the issuer processing system. If the Merchant ID is an alias, then the issuer processing system has to resolve the alias from the Alias Directory Service as described in the *Alias Directory Service Implementation Guide*. (Alias Directory from Visa is available only in select markets). If the Merchant ID is a shortened PAN then the issuer processing system derives the merchant PAN using the conversion logic explained in the section "Merchant PAN and Merchant ID" in the chapter "Mobile

Push Payment Solution and Network Processing". This merchant PAN is then included in the VisaNet OCT message as shown in the figure below.

Creating Merchant PAN from Merchant ID



Issuer VisaNet Transaction Processing

To process transactions as per Visa specifications the issuer processing system utilizes data from various sources:

- Captured from the merchant information panel
- Consumer key entry on the mobile interface
- From message fields in the VisaNet message
- From information stored in the issuer processing system

Some of the key fields required in the VisaNet message, the data source for the field and description of the data are highlighted in this section. This section is only a summary – more details are available in the *Mobile Push Payment - Technical Specifications* document.

MPP Key Data Elements

	Data Name	Data Source	Description
1	Destination PAN (Field 2 – primary account number)	Merchant information panel	This is a 16-digit PAN that defines the destination address of the recipient of push payment. It is captured in the mobile interface by the consumer through key entry, QR code, NFC, or other means. This field is required in all transactions in the program.
2	Total Amount (Field 4 – transaction amount)	Merchant information panel or Consumer key entry	This is the amount for which the transaction is processed and is either key entered by consumer (e.g., in case of static QR code) or captured by mobile app (e.g., in case of dynamic QR code). The value in this field includes transaction amount and the tip amount or convenience fees if applicable. In cross border merchant transaction, the issuer is required to convert this amount into the merchant currency before processing. This field is required in all transactions in the program.
3	Tip or Convenience fee (where allowed) (Field 28 - Transaction fee amount)	Merchant information panel or Consumer key entry	If a transaction involves convenience fee or tip, issuers are required to populate this field with the tip amount or convenience fee amount as key entered by consumer or obtained from merchant information panel. In cross border merchant transaction, the issuer is required to convert this amount into the merchant currency before processing. This field is applicable only for merchant payment transaction.
4	Merchant name (Field 43 - Card Acceptor Name/ Location)	Merchant Information Panel or Response message from acquirer	This contains the name of the merchant as available in acquirer system. Issuer is recommended to use this data to populate consumer digital receipt and statement narration, instead of the information as available on merchant information display. This field is applicable only for merchant payment and cash-out transaction.
5	Additional Data Field – Value 1 (Field 62.7 – Purchase identifier)	Merchant information panel or Consumer key entry	This is the first field which the merchant or acquirer may use to obtain additional data along with the payment transaction amount to process the request. Example of data in this field includes order tracking number for online delivery, invoice number for a store and mobile number for pre-paid top-up. The data is the field obtained from the merchant information panel or it can be key enter on mobile interface. This field is applicable only for merchant payment transaction.

MPP Key Data Elements

	Data Name	Data Source	Description
6	Additional Data Field – Value 2 (Field 102 Account Identification 1)	Merchant information panel or Consumer key entry	This is the second field which the merchant or acquirer may use to obtain additional data along with the payment transaction. Example of data in this field includes consumer identification number. The data is the field obtained from the merchant information panel or it can be key enter on mobile interface. This field is applicable only for merchant payment transaction.
7	Type of payment (Field – Business application identifier (BAI))	Issuer system	This field should be populated with the business application identifier (BAI) which defines the purpose of the transaction: merchant payment, person-to-person transfer, cash deposit or withdrawal. This field is required in all transactions in the program.
8	Consumer PAN (Field 104 - Sender account Number)	Issuer system	This is the consumer's PAN (i.e. sender PAN) available in issuer PAN management system. VisaNet uses this value as one of the inputs for interchange reimbursement fee (IRF) calculation. This field is applicable only for merchant payment and cash-out transaction.
9	Consumer name (Field 104 - Sender Name)	Issuer system	This is the consumer name as available in issuer system. This field is applicable for merchant payment and cash-out transaction. In case of cash-in, the data in the field is the merchant name.

Functional Components of the Issuer System

The capabilities required by the issuer in the program are an extension of their existing payment infrastructure.

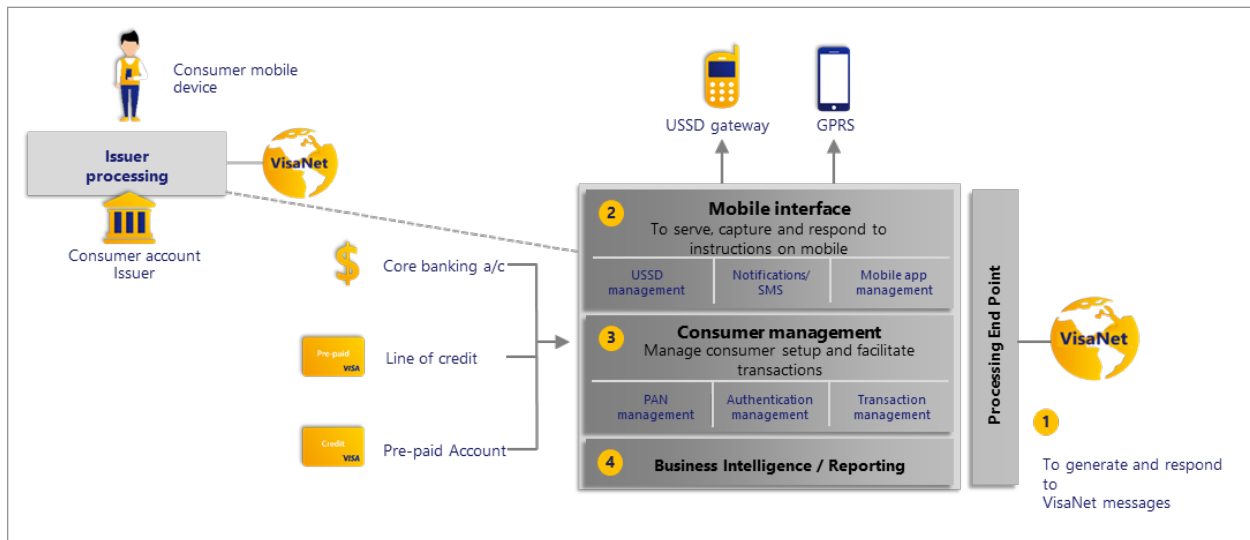
They typically reside in a technical environment similar to the issuer's card, ATM, POS, branch, or internet banking service offerings.

The required capabilities can be grouped under five distinct **components** described below. These components are not guidance on the technical design of the processing system, but rather are grouped together to provide an indication of the functional capabilities required from it.

1. **Switch (or Processing end point):** capability required to process transactions with VisaNet
2. **Mobile Interface:** capability required to process consumer interaction on the mobile and capture of data in the merchant information panel.
3. **Consumer Management:** capability required to manage the consumer account like PAN lifecycle, authentication, KYC, and account maintenance.
4. **Business Intelligence and Reporting:** capability required for account history, analysis, and tracking.
5. **Store of Value:** Linkage with the consumer account.

The components are depicted visually in the figure below and described in the table below

Issuer Processing



Issuer Processing System Capability

Feature		System Capability
Component 1: Processing End point		
1	VisaNet message processing	Generate SMS (single message) Original Credit Transaction (OCT) as per specifications (API/ISO). Place data in fields as specified. • Manage exception processing e.g., disputes, sales refunds. • Block reversals. • Support forex conversion in cross border transactions.
2	Incoming messages from VisaNet	Evaluate incoming messages, determine appropriate account to route to, and perform required action.

Issuer Processing System Capability

Feature		System Capability
3	Clearing and Settlement	Pass instant debit or credit entries to the account based on VisaNet message generated or received.
4		For settlement requests, capture full settlement details including the settlement entities, PAN, transaction type and amount.
5		Provide reconciliation or net settlement reports.

Component 2: Mobile Interface

6	Mobile app management	- Provision and control of mobile applications. - Drive logic and sequence of application data input prompts. - Support different technologies like QR, NFC and BLE, for merchant data capture e.g., QR code parser for scanning QR code.
7	USSD session management	- Connect with MNO USSD gateway. - Drive logic and sequence of the menu presentation. - Multilingual capability of the USSD menu. - Manage USSD session timing.
8	Digital receipt to consumer mobile	- Initiate digital receipt as payment confirmation based on the transaction performed. Extract required information from the VisaNet message and populate in receipt. - Send SMS receipts where applicable to feature phones.

Component 3a: PAN Management

9	PAN mapping	Assign PAN on consumer on-boarding - existing PAN on card if it exists or create PAN. Map underlying account to the PAN. Follow Issuing (ISO) BIN rules and BIN management if creating PAN.
10		Manage PAN lifecycle.
11		Support ability for one-to-many and many-to-one linkages between device, PAN, and account.
12		For outgoing transactions, based on user input and defined specifications, generate PAN.

Component 3b: Authentication Management

13	User Authentication	Authenticate the user for each transaction by checking the device ID or MSISDN of the mobile device used or other risk-based authentication.
14	Authentication credentials	Authenticate PIN/biometric for each transaction initiated by the user. Manage changes.

Component 3c: Transaction Management

15	Account lifecycle support	Enable consumer onboarding. Manage account lifecycle – create, activate, suspend, and close.
16	Transaction routing	Manage transaction processing amongst different components of the system like switch, mobile interface, and account.

Issuer Processing System Capability

Feature		System Capability
17	Fees	Manage consumer fees setup and calculation.
18	Back office management	Ability to create back office users and maintain their access rights and control.
19	KYC maintenance	• Maintain KYC records of user segments where accounts are held in processing system and not external in Core banking. • Support KYC validation and AML check for recipient.
20	Reconciliation and Financial book-keeping	Provide information for reconciliation of financial transactions.

Component 4: Business Intelligence and Reporting

21	Transaction history and user data	Facilitate customer care by providing required information on customer activity and their status and maintain required database.
22	Risk management and control	• Provide risk analytics for identifying fraudulent activity. • Provide fraud prevention, detection, and control.
23	Exception/ Dispute Management	• Provide exception handling capabilities and reports. • Provide operational requirements like reconciliation.
24	MIS, Reporting requirement	Provide reports for day-to-day operational management of the client program – for finance, business intelligence, operations, regulatory, front office.

Component 5: Store of value

25	Store of value	Create linkages with account, line of credit or pre-paid store of value.
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Configurations of the Issuer System

The processing system components required by the issuer to deliver push payment capabilities may already be available or can be created in-house or may be outsourced.

The issuer should consider their existing technical capabilities, business requirements and time-to-market in deciding the configuration of their processing system. This section describes three possible configurations of the processing system that issuers can select from. Architecture diagrams for these three configurations are available in the *Mobile Push Payment - Technical Specifications* document.

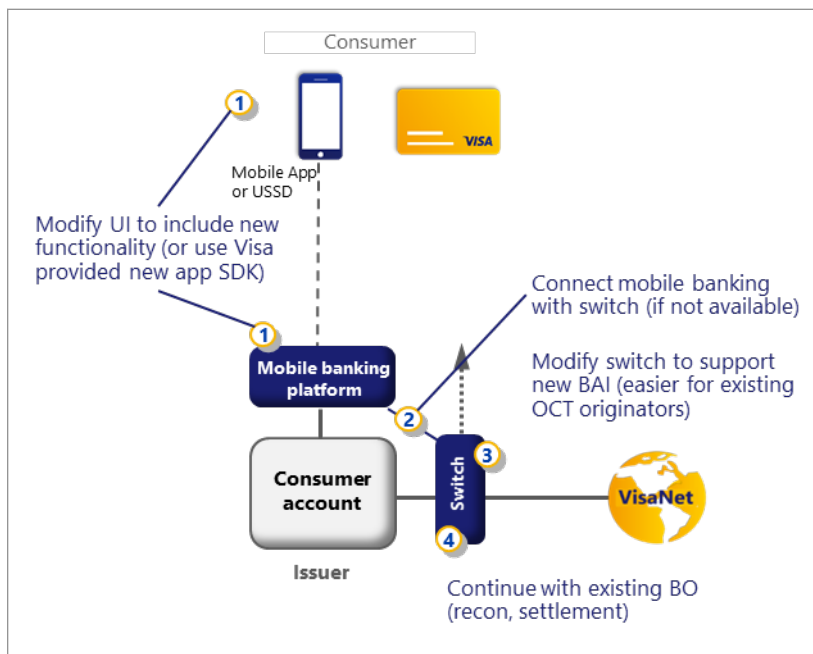
Option 1: Issuer Manages the Entire System in-House

In option 1, the issuer selects to manage the entire processing in-house.

The following changes may need to be made by the issuer, as reflected in the figure below.

1. Modify (or create) the mobile user interface to include Visa push payment transactions.
2. Connect the mobile banking platform to the existing switch (if not already be in place).
3. Make changes in the switch to support push payment transactions.
4. Put new processes in place to continue using the back-office functions of the card system - including the reconciliation and settlement functionality.

Option 1 Issuer System Configurations

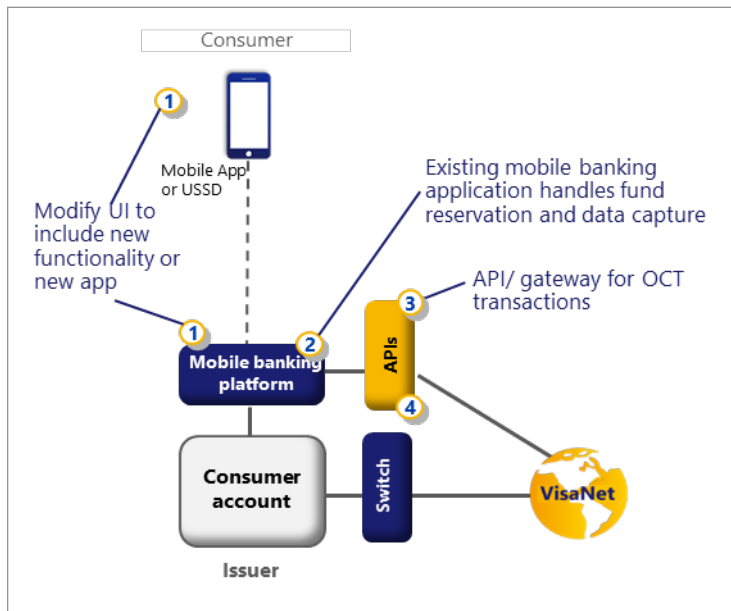


Option 2: Client using Visa API for OCT Transactions

In option 2, the in-house ISO8583 processing switch is not used to process VisaNet transactions.

Instead, Visa API with linkages to VisaNet handles the transaction processing as shown in the figure below. The issuer has to modify (or create) the user interface for the mobile device to include push payment transactions. The existing in-house mobile banking application handles the other processing requirements like fund reservation and data capture. Refer to developer.visa.com for integration details.

Option 2 Issuer System Configuration



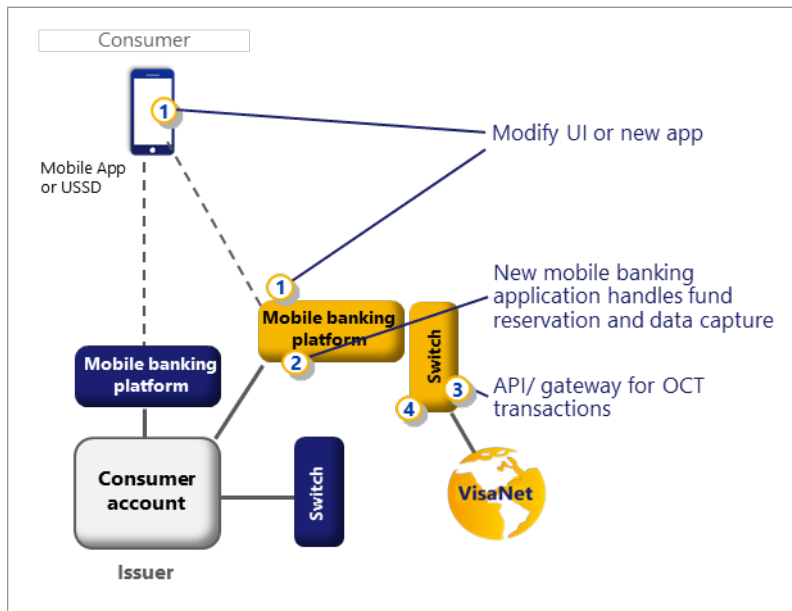
Option 3: Client using Third Party Processing System to run the Mobile Banking Platform

In option 3, an outsourced provider manages the mobile banking application, the transaction processing and the connection to VisaNet as shown in the figure below.

The mobile menus are managed via the outsourced system as is the VisaNet processing.

1. Incorporate the Visa transaction menu in the issuer's existing app or create a new app.
2. Connect to an outsourced platform that handles fund reservation and data capture
3. The outsourced provider has the gateway for Mobile Push Payment transactions.

Option 3 Issuer System Configuration

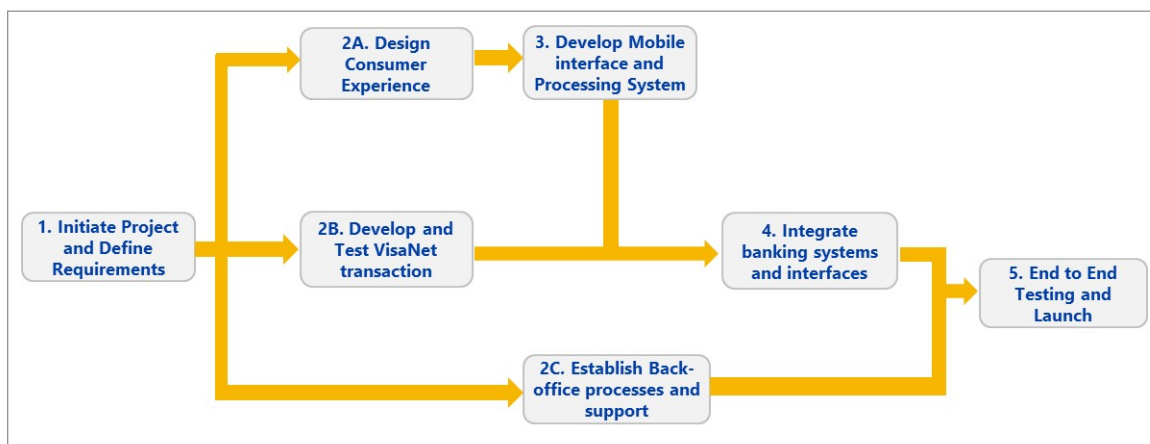


Issuer Implementation Checklist

This section provides indicative implementation checklist for the issuer.

It is for informational and reference purposes only, and should not be relied upon for operational, legal, technical, financial, or other advice. The checklist has been divided into typical stages of the issuer implementation process. Typical stages in the issuer implementation project for deploying the solution is depicted in the figure below.

Issuer Implementation Project - Typical Stages



| Stage 1: Initiate Project and Define Requirements

Project Initiation

- Obtain executive sponsorship for the project.
- Identify project leader and project manager.
- Constitute core project team, which has representation from different internal departments.
- Seek regulatory approval as required.

Create and Publish Project Plan

- Conduct workshop with Visa to understand detail requirement.
- Refer Issuer implementation overview document from Visa.
- Decide on the optional transactions to be included.
- Create a project implementation plan and agree on launch timelines with involved departments.
- Decide the type of accounts that will be included in the program.

Define Business Requirement

- Create a business requirement document (BRD).
- Refer Visa provide business requirement document template.
- Validate if the BRD is compliant to program requirement.
- Submit Program Registration form

Define Technical Requirement

- Determine high level system requirements
- Conduct gap analysis against existing capabilities.
- Evaluate mobile interface option – SDK versus app development.
- Compare effort required for API connection with ISO connection to Visa.
- Decide whether the system will be in-house or outsourced.
- If outsourced, conduct partner evaluation and selection.
- Finalize processing system configuration details.
- Seek budgetary approval.
- Issue purchase order to outsourced partner.
- Initiate project with Visa Implementation Manager.

| Stage 2A: Design Consumer Experience

Determine Mobile Interface

- Decide on the mobile channel – mobile app and/or USSD.
- Decide whether to include proposition in existing app or create a standalone app.
- In case of more than one consumer app, decide on the appropriate app to deploy the solution.

Design Mobile Interface

- Conduct session with Visa design team to identify most appropriate user experience.
- Plan to include different use case scenario such as tips, convenience fee, additional data etc.
- Submit wireframe of experience to Visa before starting development.
- Submit user experience to Visa for approval.

Determine Authentication Methods

- Determine the appropriate mobile channel authentication method.
- Adopt PIN or biometric or other risk-based authentication for a fast transaction experience.

Design Consumer Onboarding process

- Establish process to link Visa PAN to underlying account.
- Define process to onboard consumer with existing Visa card and without Visa card.

Finalize Fees and Limits

- Determine the transaction limit and overall limit on the account.
- Establish the appropriate reward and loyalty program.
- Define rewards program for consumer
- Finalize any consumer fees for applicable transaction.

| Stage 2B: Develop and Test VisaNet Transactions

Submit Project Forms to Visa

- Submit required forms to Visa Implementation Manager.
- New BIN application may be required
- Obtain detailed project plan from implementation manager.
- Determine the VCMS test environment availability date.

Develop VisaNet/API transaction

- Develop Visa messaging capability through ISO or API.
- Plan for exception scenarios including advice message.
- Refer Visa test script and prepare for testing.

Test QR code

- Use Visa provided QR code for validation of app.
- Submit the results to Implementation Manager before testing online message with Visa.

Conduct VisaNet Testing

- Prepare for end to end testing starting from mobile interface to VisaNet.
- Conduct self (offline) testing with VCMS.
- Conduct attended (online) testing with Visa.
- Validate back office settlement and reconciliation based on the sample online transaction.
- Communicate production cut over steps and timelines to Visa.

| Stage 3: Develop Mobile Interface and Processing System

Develop mobile interface

- Determine and publish app support on various mobile operating systems
- Include SMS as well as in-app digital receipts.
- Digital receipts should be initiated only after response from acquirer.
- Receipt should carry merchant name and location as received from acquirer.
- Finalize all other relevant content of digital receipt.
- Share content of digital receipt with Visa as part of app review.
- Develop mobile interface, USSD or app.
- Validate app development against different QR code test guide of Visa.
- Share results with Visa.
- Establish and sign off mobile channel security requirement.
- Implement mobile channel authentication policy.
- Finalize and test mobile interface with backend.

Develop the Mobile Processing System

- Develop the processing system as per the requirement of the program.
- Consumer account should be debited only after receiving success response from Acquirer.
- Decline transaction should result in immediate reversal of funds.

- Validate system capability with product features.
- Integrate app or USSD with the backend system.
- Develop meaningful success as well as failure notification capability.
- Develop cross border transaction processing capability.
- Determine forex rates and markups for cross border transactions.
- Develop system configuration against PCI-DSS security requirement.
- Develop reporting requirement of different stakeholders.

| Stage 4: Integrate Processing Systems and Interfaces

Develop Integration between Systems

- Finalize specs for the interface between mobile app and mobile processing system.
- Integrate and test the interface between mobile app and mobile processing system.
- Develop interface between mobile channel backend system and VisaNet gateway.
- Plan for exception scenarios in transaction processing.
- Develop mobile processing system and backend host (store of value) interface.
- Conduct System Integration Testing.

| Stage 2C: Establish Back Office Processes and Support

Back Office Processes

- Establish a process of dispute handling.
 - Create dispute-handling document.
 - Obtain access to Visa Resolve Online (VROL) and identify internal processes.
- Determine the various reports that would be required.
 - Create regular report list and frequency.
 - Finalize procedure for report delivery and analysis.
- Determine call center support required for customers.
 - Develop policies and procedures.
 - Identify point of contact for consumer queries.
 - Create quality standards.
 - Develop procedures for service enquiries.
 - Create customer service scripts.
 - Create welcome calling scripts and plan processes to make it operational.

- Identify settlement and reconciliation requirement and make back office system changes.
- Work with Finance and Accounting department to determine different ledger and income accounts.

Audit, Fraud and Risk Management

- Determine system controls and business processes to monitor fraud levels and manage the overall program risk.
- Conduct system IT security testing and penetration testing.
- Conduct technology risk assessment and establish controls.
- Determine and document ongoing Audit requirements and control for program.

Training

- Identify training needs and timelines for core project team, back office staff, call center staff and Branch staff.
- Develop training objectives and requirements as per audience.
- Develop training materials.
- Produce operational manuals and help guides.
- Produce internal departmental training schedule.

Legal Activities

- Finalize legal contracts and get internal legal approval.
- Sign all contracts or publish approved contracts.
- Obtain regulatory approval as required under prevailing regulation, if any.

| Stage 5: End-to-End Testing and Launch

User Acceptance Testing

- Processing system testing.
- Internal user acceptance testing.
- Staff testing.
- Friends and family testing.

Launch Planning and Market Launch

- Launch the service to all consumers backed with marketing activities.
 - Develop a high-level marketing plan in accordance to the objectives and business plan.
 - Develop consumer education plan.
 - Define the public relations strategy for launch announcement.
 - Define promotional and advertising strategies.
 - Appoint and brief promotional and advertising agencies, if required.
 - Identify key benefits to highlight in the marketing campaigns.
 - Ensure targeting and message content is consistent with strategic objectives.
 - Define the marketing channels to be used and the requirements.
 - Design promotional and advertising materials.
 - Obtain legal approval for materials.
 - Produce and distribute promotional materials.

Chapter 4

Acquirer Processes and Processing System

This section describes the business processes and processing system capabilities that are required by an acquirer in the program.

It explains various options for merchant information display and merchant notification, which combine to serve different payment use cases. Processes like merchant onboarding are described. Components and possible configurations of the acquirer processing system are explained.

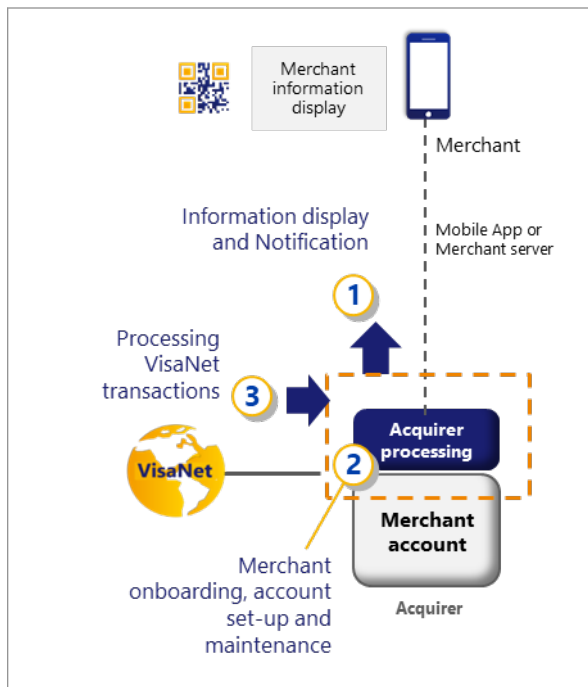
Functionality Required by Merchant Acquirer

To participate in the program, an acquirer has to deploy business processes and processing capabilities that can be grouped under three broad areas.

These are depicted in the figure below.

- 1. Merchant information display and merchant notification:** The acquirer has to provide a “merchant information display” to the merchant. This display contains information about the merchant, required by the consumer to initiate the payment and for VisaNet to process it. The acquirer has to notify the merchant instantly with a digital receipt for the transaction. The information display and notification method are of different types based on the payment use case.
- 2. Merchant onboarding, account set up and maintenance:** The acquirer has to create the merchant PAN and map it to the merchant account in their processing system. The merchant PAN, and the associated Merchant ID, is included in the merchant information display. The merchant notification method is mapped to it. The acquirer system manages features of the merchant account through its lifecycle.
- 3. Processing of VisaNet transactions:** Acquirer’s processing system has to receive, respond to, and act on VisaNet messages and manage exception scenarios.

Three Broad Acquirer Processing Capabilities



Merchant Information Display and Merchant Notification

The acquirer deploys two key capabilities in the program: provision of the **merchant information display** for use by consumers to transact conveniently at the merchant, and deployment of reliable **merchant notificationchannel** to inform the merchant of the status of the payment being made to them.

Merchant Information Display

The merchant information display consists of information about the merchant (and optionally about the payment being presented) in a format that is conveniently “captured” by the consumer using the mobile channel.

- The acquirer has to ensure that brand guidelines as referenced in the section "[Branding Standards](#)" in the chapter "Mobile Push Payment Solution and Network Processing" are followed in the merchant information display.
- Specifications for the merchant information to be included in the merchant display are described in the *Mobile Push Payment - Technical Specifications* document. Acquirers should follow the specifications carefully and test the merchant display for errors before deployment, to avoid transaction failure. Visa provides online tool for validation of merchant information display.
- The merchant information display can be of various types. It can be a static or dynamic QR code, NFC tag, Bluetooth beacon, deep-linked app-to-app trigger or simply the numeric or alphabetical Merchant ID. Refer to the earlier section "[Merchant Information Display and Notification](#)" in the chapter "Mobile Push Payment Solution and Network Processing" for more information. It is mandatory for the merchant display to include the QR code in all location, and the Merchant ID where applicable.

The type of merchant display deployed is determined by use case, the type of payment transaction (e.g., remote, or face-to-face) and by the optional information the merchant wants to present (e.g., invoice number and payment amount). Some examples are described below.

- **Face-to-face payments:** The merchant information is presented on a decal displayed by the merchant at the point of transaction. If the point of transaction is fixed, the decal is placed at the point of transaction (e.g., countertop, tabletop, or vending machine). If the point of transaction is flexible, the decal is carried by the person conducting the transaction. (e.g., on a lanyard or on a plastic card). If the merchant presents dynamic data (e.g., amount to be paid, merchant invoice number), the display has to be dynamic as well. This dynamic QR code can be shown on a merchant app, a display screen of the cash register, or printed on the generated invoice. The data can also be presented through NFC or Bluetooth transmitting devices or any other new technology in the future.
- **Remote payments:** The merchant information is displayed on the artifact used when conducting the transaction. This artifact can be a bill, an invoice, a poster, checkout page of the online merchant, email, or an online message. The display can be in the form of a QR code or can be a message from the merchant to the consumer containing a link that triggers the consumer's mobile app to open on its payment screen. The information presented is dynamically created by the merchant, using the specifications as described in the *Mobile Push Payment - Technical Specifications* document.

Merchant Notification Channel

The merchant notification channel is used to inform the merchant of the status of the payment being made to them.

Acquirer can deploy various notification channels based on merchant need. The most common notification channels were listed earlier in the section "Merchant Information Display and Notification" in the chapter "Mobile Push Payment Solution and Network Processing" and are described below. The acquirer can also deploy other innovative and reliable notification channels.

- **SMS message:** The notification is in the form of a mobile SMS message to the merchant's mobile device. However, SMS message delivery can be unreliable, and the message should have safeguards against fraudulent behavior such as spoofing.
- **In-app notification:** The notification is an in-app message to the merchant's mobile app provided by the acquirer. This method is more reliable than the SMS message. The notification can be stand alone or can check-off an invoice created using the app (e.g., with the invoice number included in the QR code created by the app).

- **Notification to an IP POS device:** The notification is to an existing IP POS device that can display the message on the POS device or even trigger a charge slip to be printed.
- **Notification to a server, or to a cash register system:** The notification is to a merchant server where it can check-off an invoice already created in the server (e.g., utility bill) or trigger an action (dispense from an automated vending machine). The notification can also be routed to an existing ERP system or integrated billing system (cash register) deployed by the merchant to trigger a message at a particular payment counter (e.g., the payment lane where the customer is making the payment in a multi-lane environment).

Notification to a server can be delivered through APIs provided by the acquirer system. The APIs should consider the following: easy integration with the merchant system; different technologies adopted by the target segment; security requirements; different failure scenarios; and tools to help merchant conduct day-to-day operations and reconciliation.

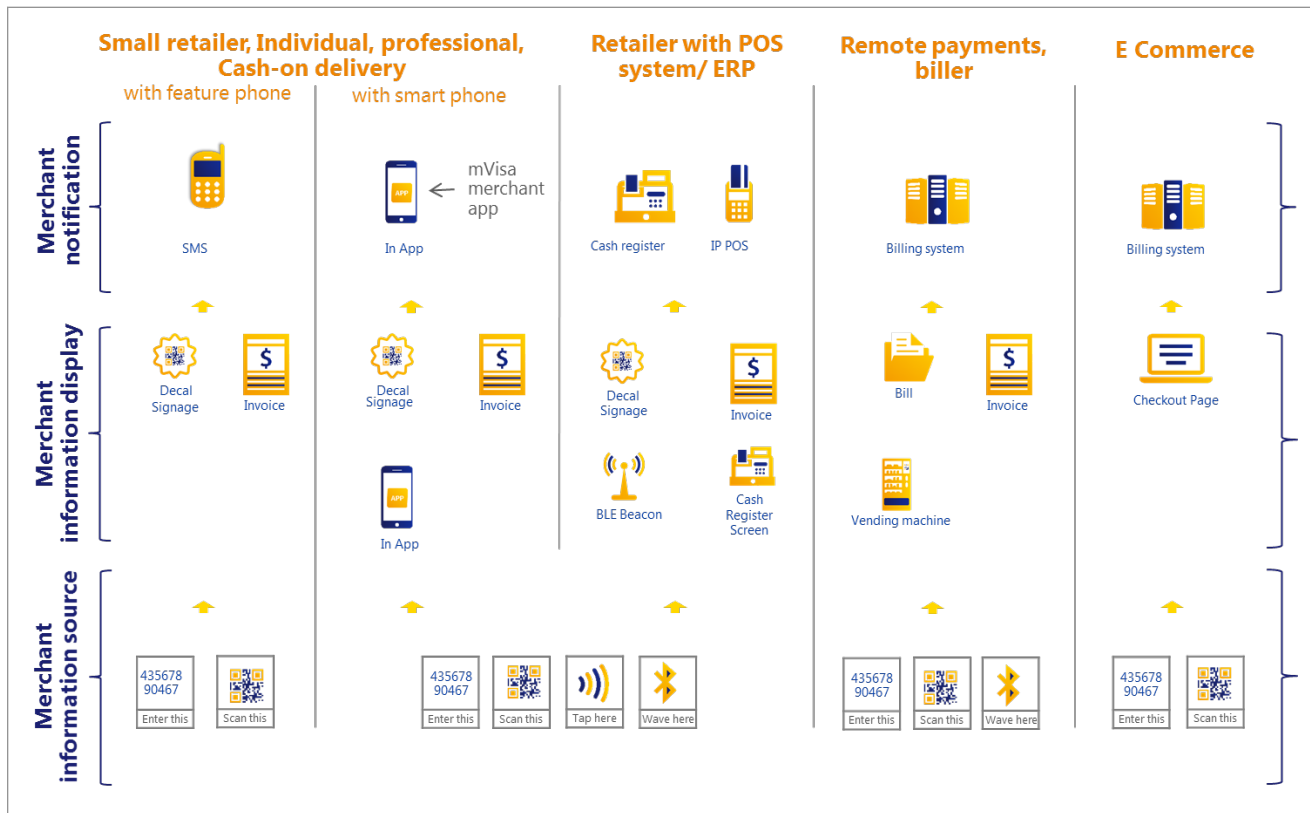
- **Email notifications:** The notification is to an email ID provided by the merchant.

Combination of Merchant Information Display and Notification Channel

The combination of merchant display and notification channel deployed at a merchant is influenced by various factors - the payment use case, merchant preference, type of merchant business and the size of merchant establishment.

Some such combinations are illustrated in the figure below.

Merchant Display and Notification Channel Combinations



Merchant App and its User Experience

The merchant app is an important acceptance tool from the acquirer, especially for targeting the small and medium size merchant segment.

The merchant downloads the app and the merchant account is linked with it. Acquirers will have to select the app platforms through which they will make their app available in the market. The acquirer should utilize the mobile SDK provided by Visa to simplify the implementation.

When designing and creating the merchant app, the acquirer should keep the following in mind.

- The user experience and **mobile screens** to be used for the merchant app have been defined in the *Mobile Push Payment – Mobile User Experience Guide* document.
- Acquirers can modify the user experience and screens within the guidelines. Any modification in the mobile screens should be designed with the objective of simplifying the transaction experience for the merchant. The look, feel and colors of the screens can also be modified to comply with the acquirer's brand guidelines. The final screens and user experience should be submitted to Visa for approval before the app is created.
- The app must have the ability to generate a "QR code" for the consumer to scan, based on input from the merchant.
- The app should provide for varied access rights by merchant personnel as explained further in section "Access Options for the Merchant Account" in this chapter.

Additional Services via the Merchant App

The merchant app is software led and cloud based. It presents opportunities for the acquirer to expand their business relationship with the merchant in ways that are otherwise challenging with a hardware bound solution.

The features built into the app by the acquirer can deliver benefit to the merchant and additional business for the acquirer. Features that acquirers can consider including in the app are:

- **Short-term working capital loans:** Merchant builds transaction history with the app, which may entitle the merchant to working capital loans from the acquirer. The app can act as a quick access to urgent funding needs and as a repayment mechanism. (e.g., merchants view loan amount eligibility that changes with their transaction behavior.) Any loan availed is credited to the merchant account and transactions accepted are routed towards loan repayment.
- **Business intelligence:** Small and medium size merchant may not have the tools to get real time access to sales data or to derive business intelligence from historical data. Acquirers could provide the merchant access to business intelligence that can be leveraged by the merchant to improve their business. (e.g., information of when the footfall is the highest during the day so that manpower can be planned, and customers do not go unserved).
- **Merchant loyalty program:** Small and medium size businesses may want to offer loyalty programs to their customer but do not have the ability to manage such a program. The acquirer can create a merchant loyalty program which can be configured and administered by the merchant using the mobile app. (e.g., reward the consumer for every fifth purchase)

Merchant Onboarding

Merchant onboarding is a key business process for the acquirer with which to expand their acceptance footprint and leverage the advantages of the program.

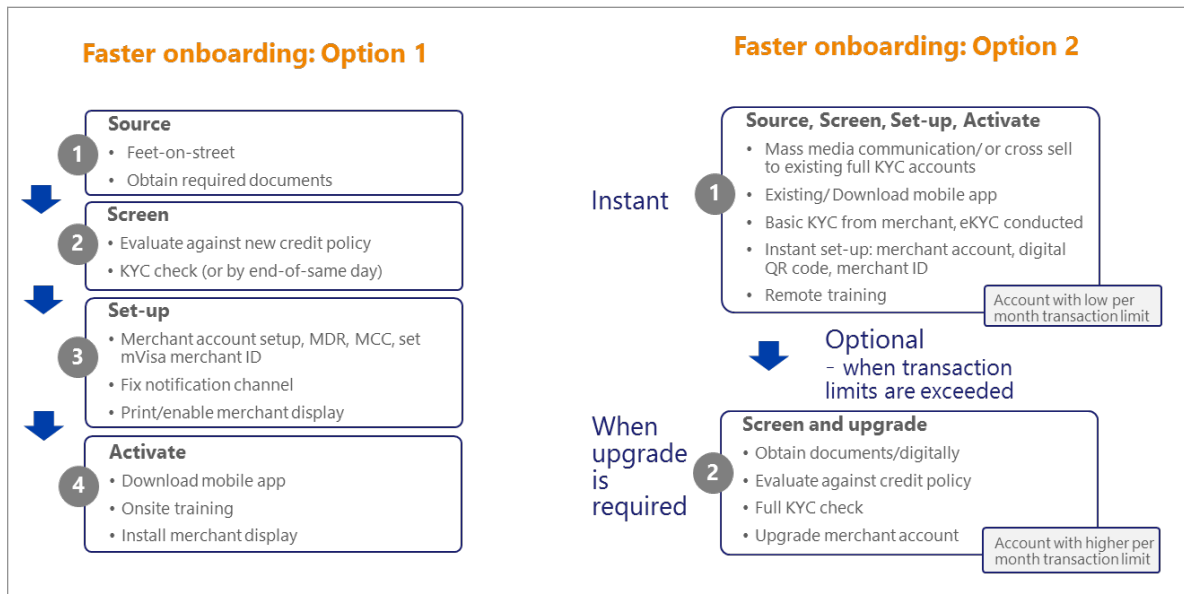
In the regular acquiring business, the merchant experience during onboarding can be onerous - involving submission of extensive documentation to cover for the associated business risks. Since the program is based on push transactions that involve lower risk, the acquirer should revisit their existing processes and modify them for the solution where required.

Merchant Onboarding Process and Time

The merchant onboarding process consists of sourcing, screening, set-up and, activation of merchants.

The typical process will consist of the steps outlined in the figure below. The acquirer should evaluate all the steps to create an end-to-end process that is instant or takes the least amount of time and which can be further improved over time.

Merchant Onboarding Process



Merchant Sourcing and Targeting

The acquirer should identify target segments from which they will source merchant accounts for acceptance.

Prospects that were earlier uneconomical to serve can now be targeted with the solution. The acquirer should use criteria like minimum business volume, geographic location, current penetration levels and accessibility in creating the prospect database.

- **Cross sell:** The acquirer may already have a significant number of business accounts in their portfolio that are small and medium enterprises. These **SME accounts** can be a prime cross-sell opportunity for the acquirer. These accounts fulfill KYC requirements and the acquirer will have internal data on the performance of these accounts. This gives an opportunity to the acquirer to pre-register the account and then adopt mass communication means to obtain the consent of the account holder.
- **Mass-marketing route:** The acquirer can adopt mass marketing tools to expand acquiring. If permitted by local regulations, **digital and online marketing** and telemarketing should be used for communication by the acquirer to reduce the cost of merchant sign-up.

The acquirer should identify and target the many segments of businesses in their markets that have very low penetration of electronic acceptance. The acquirer should also target high footfall, well-known merchant brands or holdouts to build awareness about the solution.

Screening and Underwriting Policy

The acquirer must review the underwriting process to reduce documentation required and reduce turnaround time for merchant onboarding.

Pre-approval can be considered and instant eKYC conducted using online capabilities increasingly available in markets. Since consumer initiates the merchant payment, the fraud related dispute is significantly reduced for an acquirer. This opens an opportunity for acquirers to review their merchant underwriting policy and relax its norms to support faster merchant acceptance growth.

Merchant Account Set-Up

To set up a merchant for this solution, the acquirer assigns a PAN to the merchant's account and creates the Merchant ID from the PAN.

Next, the merchant PAN is mapped to the notification device; and the notification process is set up for the device. The acquirer also provides the merchant with a merchant display (e.g., merchant decal) or sets up the process by which the merchant will create the display (e.g., specifications to print the QR code on a bill).

| Issuing (ISO) BIN Requirement and PAN Creation

| The acquirer should use a new Issuing (ISO) BIN to generate the merchant PAN.

Note: Mobile Push Payment is not included in scope for the 8 digit BIN due to the construct of Mobile Push Payment usage. The merchant account is assigned under an issuing BIN but the account is not a card/credential.

VisaNet will decline any merchant payment or cash-out transaction to a PAN outside the range setup on VisaNet by the acquirer.

The acquirer assigns and links a PAN to the merchant account during merchant setup. The Merchant ID is created by the acquirer from the PAN. The relationship between the Merchant ID and merchant PAN is described in the section "Merchant Payment - Transaction Experience" in the chapter "Mobile Push Payment Solution and Network Processing". The acquiring processing system should be capable of generating merchant PAN, assigning it to a merchant account and deriving the Merchant ID for it.

Merchant Information Display

The acquirer has to provide the merchant with a suitable merchant information display that contains the merchant PAN and Merchant ID.

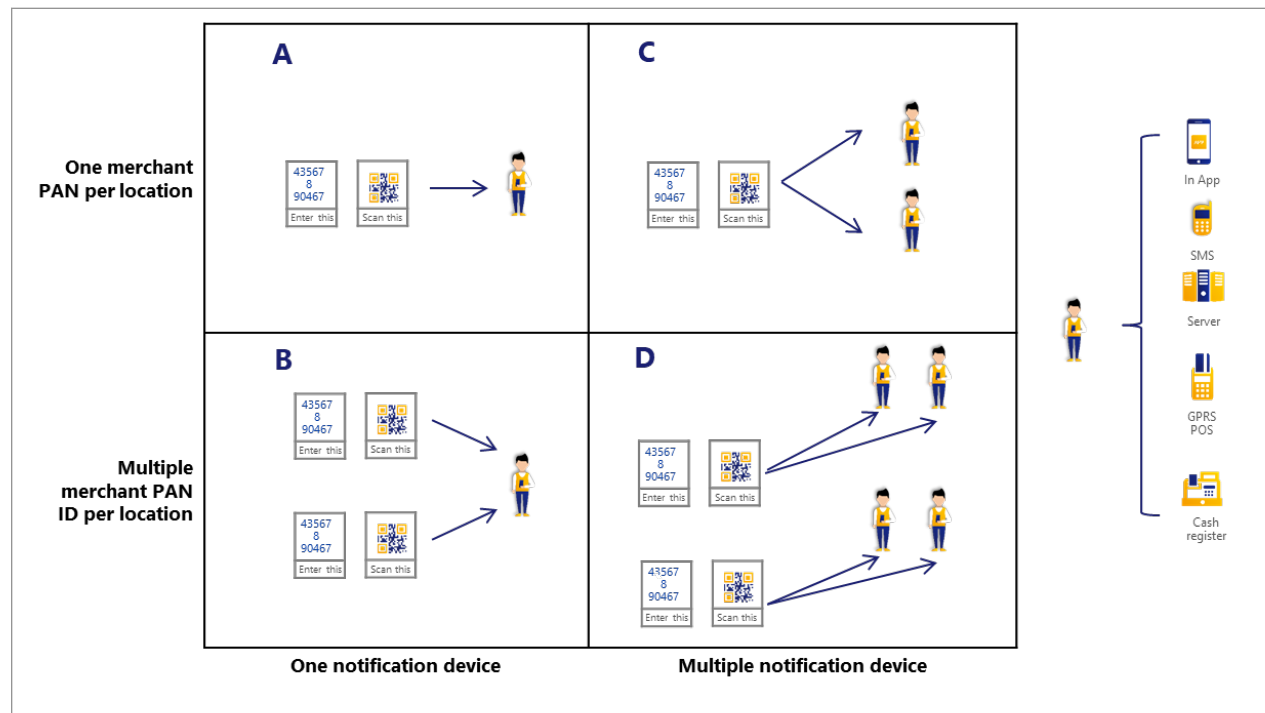
The display type is determined by the acquirer's own assessment of the most suitable option and the preference of the merchant. However, all merchants will have to display at a minimum the static QR code. Where relevant, a Merchant ID should be included. The acquirer will have to establish a process for printing or creation of the merchant information display for each merchant. The acquirer should follow Merchant display testing process recommended by Visa (e.g., for the QR codes) to prevent transaction errors. Tools are provided by Visa for QR code generation and testing and these should be leveraged. A welcome pack containing the merchant display and merchant training material should be delivered to the merchant.

Mapping Merchant PAN with Notification Channel

The acquiring system should be capable of handling more than one notification device for a given merchant PAN.

Similarly, notifications for multiple merchant PAN's should be capable of being routed to one common device. The figure below provides the possible combinations that the acquirers system should be capable of managing.

Mapping Merchant ID with Notification Device



Scenario A: A store is assigned one merchant PAN and given a single display and the merchant gets notifications on a single device. Example of such merchants is small grocery store, barbershop, doctor's clinic etc.

Scenario B: Here different merchant PAN is encoded in different merchant displays to identify different locations or lines of business. However, the notification for all them is directed to one device. Example of such a merchant is an entrepreneur owning two different businesses.

Scenario C: In this situation, the store has one merchant PAN. The merchant display can be one or multiple copies of the same display. However, the merchant is provided with notification on more than one device. Example of such a merchant is a restaurant with the person waiting the table as well as the cashier receiving notifications for the same payment. Another example is a small corner store with the notification being sent to the shop attendant and the shop owner.

Scenario D: Here large merchant establishments or business-to-business supply chain have combination of all the above scenarios with the multiple merchant PAN, displays and notification devices for the same merchant. Example of such a merchant is a supermarket. Another example is of FMCG delivery trucks - the delivery person as well as the warehouse receive a payment notification.

Access Options for the Merchant Account

The acquirer should provide secure access for merchants to their merchant account to process sales cancellation or refunds, to review transaction logs, track sales and any other account related data. This access can be provided over a variety of touch points such as USSD channel, merchant app or through a web portal.

The acquirer should use mobile device authentication for merchant access using USSD channel or a merchant app. Merchants may want to use the same mobile device between multiple users, each user having different access rights depending on their role in the merchant team. A sample scenario is described in the table below. Therefore, merchant access should be using login id and password, which can be used on multiple devices but with accessed functionality limited by the role of the user at the merchant.

Access Rights of a Given Role within the Merchant Establishment

Owner or senior management or Head office	Store supervisor	Frontline or customer service staff
• Complete account access • Business intelligence across multiple stores • Ability to avail line of credit	• Ability to authorize or process sales refund or reversal • Access to in-store periodic sales data • Loyalty programs administrative function	• Receive transaction notification • Review transaction log and limited transaction history

Merchant Settlement Time

Acquirers have to provide instant notification to the merchant about the status of the payment made to the merchant.

However, merchant notification and merchant settlement are different. The acquirer has to agree the terms of settlement time with the merchant - the merchant can get instant access to the funds in their account, or the funds are credited later within the agreed number of days. The acquirer gets access to the funds after the VisaNet settlement cycle, which is at least once daily. This cost of funds can be built into the merchant service fees or optionally acquirer may charge extra fees from merchants who value instant settlement credit.

Merchant Training

The acquirer should create simple online or electronic training modules that explain the acceptance procedures and the acceptance processes to avoid the cost associated with a site visit for merchant training.

Merchant Acquirer Processing Requirements

The acquirer system has to manage the features of the merchant account through its lifecycle.

The processing system has to receive, respond to, and act on VisaNet messages and manage exception scenarios.

Option to Select either Single Message or Dual Message Processing

Acquirers have the option of using ISO or API based connectivity with Visa.

When the acquirer opts for ISO connectivity, then the acquirer has the option of receiving the same OCT message from VisaNet in either the SMS for the BASE I (dual message system) format. The acquirer should select the message format based on their switch and system capability.

- If the acquirer elects to receive the transaction as a dual message, VisaNet converts the SMS format from the issuer to the BASE I format for the acquirer. The acquirer's switch has to respond online to BASE I transactions from VisaNet and later process the BASE II files received at the time of VisaNet settlement.
- If the acquirer elects to receive the transaction in SMS format, the acquirer's switch has to respond online to SMS transactions from VisaNet and later process the associated raw data settlement files received at the time of VisaNet settlement.

In both cases, the VisaNet settlement timelines and dispute management process remain the same.

Required Data from the Acquirer in the Response Message

The acquirer has to provide merchant data (like merchant name, merchant location, and merchant category code) in its response to the OCT message received from VisaNet.

This data is used at the network level to administer IRF, settlement and disputes. The *Mobile Push Payment - Technical Specifications* document provides the details of the data fields utilized by the acquirer for this purpose.

Placing Transaction-Specific Data in the Merchant Information Display

In many payment scenarios, additional information is needed by the merchant to complete the transaction or improve the user experience.

Examples of such data are subscription number for a utility bill, mobile number for a pre-paid connection, transaction identifier for an online payment or simply an invoice number. Such additional

data helps merchants process transaction for the customer, apply loyalty rewards or facilitate their own invoice-to-payment matching.

- The Visa specifications enable the merchant to encode up to two such transaction specific values in the merchant information panel. These are referred to as **Additional Data Field**. The information in the additional data field displayed in the merchant information display is then routed from consumer mobile to issuer system, carried through in the VisaNet message to the acquirer and then to the merchant.
- Alternatively, the specifications also enable such transaction specific value to be key entered by the consumer on their mobile interface. The key entered data is then routed from consumer mobile to issuer system, carried through in the VisaNet message to the acquirer and then to the merchant.

Please refer to the *Mobile Push Payment - Technical Specifications* document for more details on the Additional Data Field. A summary of the feature is also described in the figure below.

Options for handling transaction specific data

	Merchant or acquirer presented value	Consumer entered free text value
	Dynamic QR code	Static QR code
Additional Data Field – Value 1	Merchant creates a dynamic QR code containing transaction specific value (<i>Tag 62 of EMVCo specifications</i>). This value passes through issuer and acquirer system and is forwarded to merchant by acquirer, along with the transaction notification. Merchant can use this value for further processing. E.g., mobile number for telecom companies; subscriber ID for cable company, student ID for school etc.	If the merchant cannot create a dynamic QR, then they can prompt consumer to key enter data in mobile interface by encoding special character "****" in the tag. Consumer entered data would be forwarded to merchant by acquirer. As this would be mandatory value, merchant would have to guide the Consumer on the information to be enter at the point of transaction, without which the transaction will not be processed. E.g., Schools should prompt payer to enter student enrollment number in additional data field for school fee payment.
Additional Data Field – Value 2	If required, merchant can create more than one transaction specific data. Merchant can provide alphanumeric special character (ANS) data in <i>Tag 62 of EMVCo specifications</i>. This is forwarded to merchant by acquirer, along with the transaction notification. Merchant can use this value for further processing. E.g., invoice number for telecom companies, inventory number for warehouse.	Similar to Consumer entered values, merchant requiring additional data or ANS value can use this field. Merchant should encode value "****" in the tag. E.g., traffic cop should prompt payer to enter vehicle number for the payment of a fine.

Enabling Merchant Payment Transaction with Additional Charges

In certain merchant payment transactions, consumers may be required to pay an additional amount over and above the transaction amount.

One common example are payments to government agencies. Please contact your local Visa representative to know about such merchant payment scenarios in your market. The technical specifications allow for the merchant to accept such payments. The acquirer can enable such transactions in the following manner:

- The merchant information display specification provides information to trigger for additional amount. The mobile app will use this trigger to calculate the additional amount and display it to the consumer before confirming the transaction.
- In case of a key entered transaction - the consumer will have to use the tip amount prompt on the mobile app to enter the additional amount.

The issuer is required to separately identify both the total amount (i.e. transaction amount plus additional amount) and the additional amount in the VisaNet message. See *Mobile Push Payment - Technical Specifications* document for more details.

Managing Transaction Exceptions

The acquirer has to plan preventive or corrective actions for scenarios where transactions might fail, using either online means or manual intervention process.

Well-designed exception management processes will reduce disputes and contain financial losses.

As an example, the success notification to the merchant could fail due to network drop, network delay or communication failure. However, the transaction will get processed by VisaNet and by the issuer, and the success notification will reach the consumer. In this failure scenario, the merchant should be able to check the status of the transaction from their acquirer's system by reviewing last successful transaction.

The acquirer should map out possible transaction failure scenarios within their system and create processes to address them. The *Mobile Push Payment - Technical Specifications* document describes the six exception scenarios for the acquirer's connectivity with VisaNet and the plan to address them. The acquirer should map out similar exception scenarios within the system and for the system interface with the merchant.

Managing Sales Cancellations or Merchandise Return

The acquirer processing system has to provide the merchant an online facility to cancel or refund a completed sales transaction.

The acquirer processes the sales cancellation or refund for a transaction through an SMS message or through a BASE II TC06 transaction. The step-by-step process for sales cancellation is described in the appendix "[Transaction Descriptions](#)".

Visa offers API for processing merchandise return details of which are available on developer.visa.com.

Managing Reconciliation and Settlement Process

The acquirer has to perform daily transaction reconciliation by comparing amounts settled by VisaNet with the transactions performed during the day.

The acquirer process should reference the Visa Settlement Services (VSS) reports and reconcile them with transaction data from the switch logs. VSS reports provide data on all transactions (successful or exceptions) sent to or received from VisaNet, along with settlement amount and interchange values.

Enabling Risk Management Controls

Acquirer should establish value-velocity risk management control in their system over transactions received from VisaNet.

This control will mitigate risk from sudden surge in transactions at any merchant location. VisaNet maintains these limits at a recipient PAN level and if a limit is exceeded, VisaNet can forward message to the acquirer with information on the exceeded limit in VisaNet message Field 48, Usage 37, position 11.

Visa supports setting up limits as per the following parameters:

- **Geographic scope** – Separate limits for domestic or international transactions
- **Limit Category** – Count limit and/or amount limit
- **Limit Timeframe** – 1-day, 7-day rolling, or 30-day rolling

| The acquirer velocity limits are set for destination PAN at the Issuing (ISO) BIN level. For example, if
| the domestic (1-day) count limit for an Issuing (ISO) BIN is set to a maximum of five hundred
| transactions, every PAN issued with that Issuing (ISO) BIN is allowed to receive only up to five
| hundred messages per day.

These limits are communicated to Visa via the Client Information Questionnaire (CIQ). Please contact your regional Visa representative for details about the risk management services.

Acquirer VisaNet Transaction Processing

To process transactions accurately, the acquirer processing system has to utilize data from message fields in the VisaNet message. Please refer to the *Mobile Push Payment - Technical Specifications* document for details of all fields in the VisaNet message. Some of the key fields in the VisaNet message, their data source and description are summarized in table below.

Key field acquirer VisaNet Transaction Processing

	Data Name	Data Source	Description
1	Destination PAN (Field 2 – primary account number)	Acquirer assigned value when setting up the merchant. During transaction is in the incoming message from VisaNet to acquirer.	This is the 16-digit PAN, which defines the destination address of the recipient of push payment, e.g., the merchant account. It is captured in the mobile interface by the consumer through key entry, QR code scan, NFC tap or other means. This field is required in all transactions in the program.
2	Total Amount (Field 4 – transaction amount)	Consumer entered or merchant assigned value. Incoming message from VisaNet to acquirer.	This is the amount for which the transaction is processed. It is either key entered by the consumer (e.g., in case of static QR code) or captured in mobile app (e.g., in case of dynamic QR code). The value in this field includes any tip or convenience fees if applicable and the invoiced amount. For a cross border merchant payment and cash withdrawal transaction, this field will always be in the acquirers' local currency. This field is required in all transactions in the program.
3	POS Condition Code (Field 25)	Acquirer assigned value in the transaction response based on the scenario (customer present or customer not present) in which merchant is using Mobile Push Payment.	Return an appropriate value of 01 for a condition of 'Customer not present' and, 05 for a condition when 'Customer present, card not present' in the message. This change is mandatory for Recipients effective October 2020. If this value is not sent, the transaction will be treated as 'Customer not present' transaction. Originators may receive the value of 01 or 05 in the authorization response message from Visa.

Key field acquirer VisaNet Transaction Processing

	Data Name	Data Source	Description
4	Tip or Convenience fee (where allowed) (Field 28 - Transaction fee amount)	Consumer entered or merchant assigned. Incoming message from VisaNet to acquirer.	If the tip or convenience fee indicator is coded by acquirer into the merchant information panel, this field will provide the tip amount or convenience fee amount. Acquirers are required to display this amount as a separate value in the notification and digital receipt to the merchant. This field is applicable only for merchant payment transaction.
5	Merchant name (Field 43 - Card Acceptor Name/Location)	Acquirer system; displayed on merchant information panel. Acquirer provides in response message sent to VisaNet.	Recipient may replace this field with the 'doing business as' name of Merchant. If replaced then, the Visa settlement reports would contain the value provided by recipient.
6	Merchant Type (Field 18 - Merchant Category Code)	Acquirer System. Acquirer provides in the response message sent to VisaNet.	Recipient may replace this value in the response message. If provided, then it must contain a valid merchant category code. VisaNet will reject the transaction with reason code 0017 if MCC is invalid. If replaced, then Visa settlement reports would contain the value provided by recipient.
7	Merchant Verification Value (Field 62.20)		Recipients are expected to populate the Merchant Verification Value used to identify merchants that participate in a variety of programs in this field.
8	Reimbursement Attribute (Field 63.11)	Code that identifies the applicable interchange reimbursement fee for a purchase transaction	Recipients are required to provide the Reimbursement Attribute to indicate participation in special programs.
9	Additional Data Field – Value 1 (Field 62.7 – Purchase identifier)	Merchant information panel or consumer key entered. Incoming message from VisaNet to acquirer.	This is the first of the two fields which the merchant or acquirer may use to obtain additional data along with the payment transaction in order to process the request as described in the section "Placing Transaction-Specific Data in the Merchant Information Display". Acquiring system must forward this data to the merchant in the merchant notification. This field is applicable only for merchant payment transaction.
10	Additional Data Field – Value 2 (Field 102 Account Identification 1)	Merchant information panel or consumer key entered. Incoming message from VisaNet to acquirer.	This is the second of the two fields which the merchant or acquirer may use to obtain additional data about the payment transaction. Acquiring system must forward this data to the merchant in the merchant notification.

Key field acquirer VisaNet Transaction Processing

	Data Name	Data Source	Description
11	Type of payment (Field 104 – Business application identifier)	Incoming message from VisaNet to acquirer.	This field contains the BAI (Business Application Identifier) which defines the purpose of the transaction: merchant payment, person-to-person transfer, cash deposit or withdrawal. This field is required for all transactions in the program.
12	Consumer PAN (Field 104 - Sender account Number)	Incoming message from VisaNet to acquirer.	This field contains the consumer PAN (i.e. sender PAN) available in the issuer PAN management system. Acquirer must store this number for refund processing and may forward this data to merchant, if required. This field is applicable only for merchant payment and cash-out transaction.
13	Consumer name (Field 104 - Sender Name)	Incoming message from VisaNet to acquirer.	This field contains the consumer name as available in issuer system. This field is applicable for merchant payment and cash-out transaction. In case of cash-in, merchant name is replaced with this data. Acquiring system must include the information from this field for inclusion in the digital receipt.
14	Settlement Service Data (Field 119)		LAC only: This field contains settlement information unique to the processing of Visa transactions by participants in a given country. In countries where client-calculated IRF is supported, the recipient of the Mobile Push Payment transaction must send Field 119 in the 0210 response message for National Net Settlement Service (NNSS) transactions. This field is not allowed in 0200 request messages for Mobile Push Payment transactions and will be rejected if present.

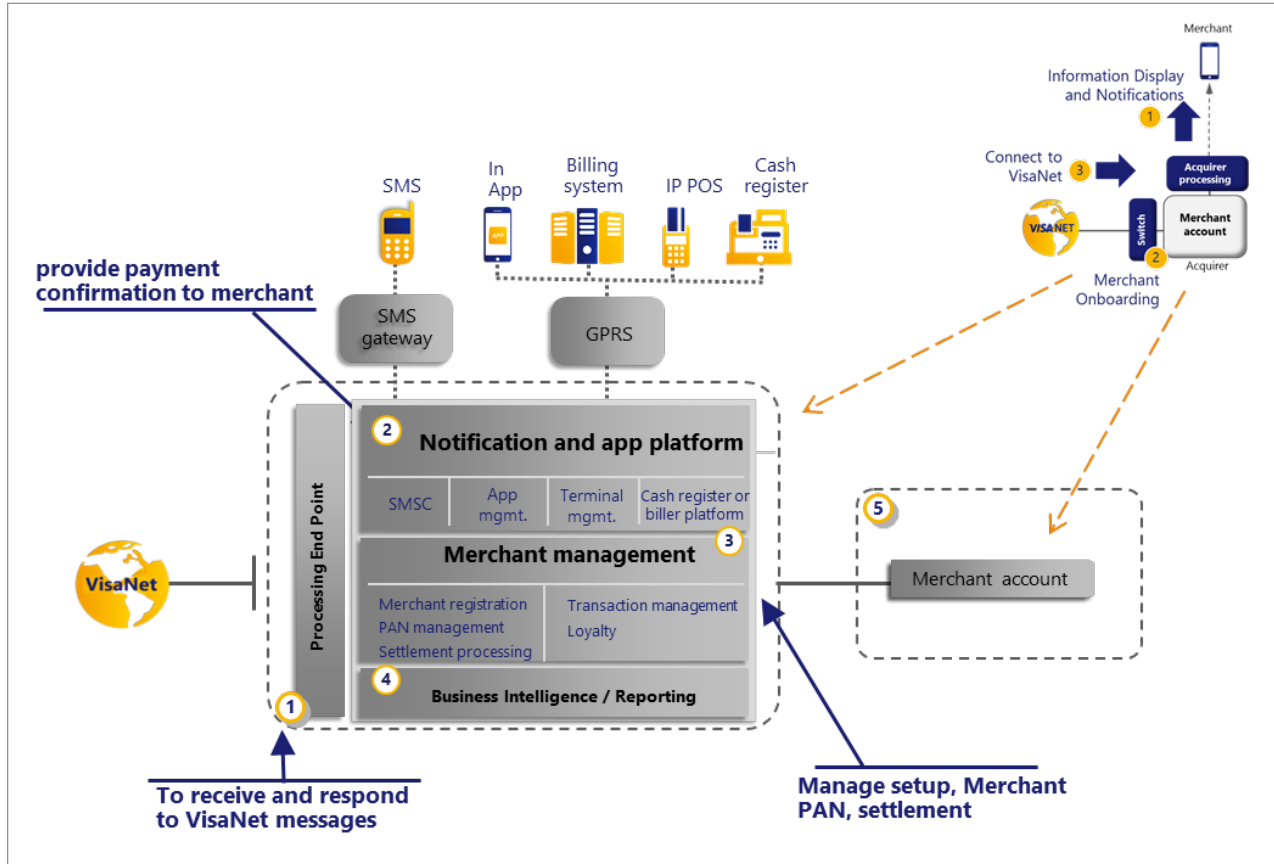
Functional Components of the Acquirer System

For an acquirer in the program, the requirements in their processing system are an extension of their card-acquiring infrastructure.

The capabilities required can be grouped under five distinct components as depicted in the figure below and as described in the table below. These components are not a guidance on the technical

design of the processing system but are grouped to provide an indication of the capabilities required from it. The acquirer has to modify their existing processing system or acquire new capability to deliver the required functionality.

Acquirer Mobile Processing System Components



Acquirer Processing System Capability

Feature	System Capability
Component 1: Processing End Point	
1 VisaNet message processing	- Receive different transaction as per specifications (API/ISO). - Handle online transaction processing exceptions like reversal advices or declines or timeout.
2 Exception, Clearing, Settlement	- Manage exception scenarios like dispute, sales refunds, merchandise return. - Provide reconciliation support and net settlement position reports.
Component 2: Notification and app platform	
3 Merchant Notification	- Enable notification to merchant devices in different scenarios. - Support integration into cash register, bill payment platform or apps.

Acquirer Processing System Capability

Feature		System Capability
4	Account Access to Merchant	• Provide merchant with secure access to process exceptions. • Enable account related access for merchant like transaction history, end of day balance, transaction status, retrieve last transaction or value-added features like loyalty administration.
Component 3: Merchant Management		
5	Merchant Setup	• Set up merchant – doing business as (DBA) name, merchant category code (MCC), merchant discount rate (MDR). • Configure merchant notification method and support to different scenarios.
6	Value added services	Rules for loyalty management functionality for merchants.
7	MDR Management	Apply merchant discount rate to transactions before crediting funds into merchant account.
8	Merchant Information Panel	• Generate merchant PAN and assign it to the merchant account. • Generate Merchant ID as per specifications for display, QR code, NFC and BLE.
Component 4: Business Intelligence and Reporting		
9	Business intelligence and reporting.	• Generate report for business intelligence, risk management, customer service, and operations management. • Provide periodic merchant statements containing relevant information like charges, amount settled, value added reports.
Component 5: Merchant Account		
10	Merchant account.	Maintain financial transaction record, credit and debits, in merchant account

Configurations of the Acquirer System

The processing system components required to deliver capabilities may already be available, can be created in-house or can be outsourced.

The acquirer should consider their existing technical capabilities, business requirements and time-to-market in deciding the configuration of their processing system. This section describes three processing system configurations that can be adopted by the acquirer. Architecture diagrams for these three configurations are available in the *Mobile Push Payments - Technical Specifications* document.

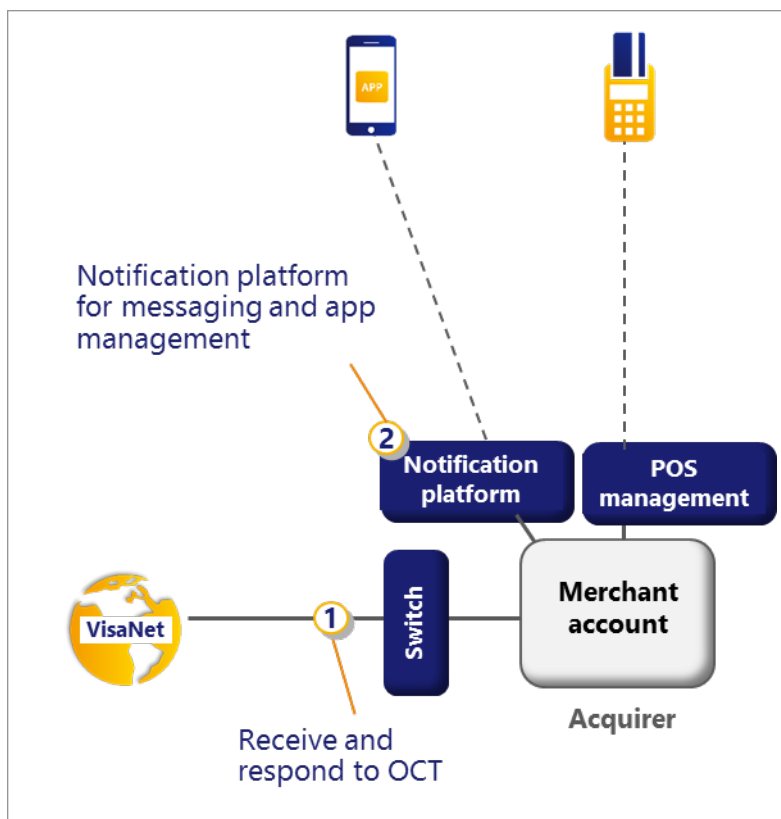
Configuration 1: Acquirer Owns System that is Managed In-House

In this option, the acquirer manages the entire acquiring processing system in-house as depicted in the figure below.

The following changes may need to be made in an already deployed system:

1. Inclusion of a notification platform to enable notification messages to be delivered to the merchant and the app management system.
2. Changes in the financial switch in order to support different transaction – receive and respond to OCT transactions.
3. Inclusion of usual processes to continue using functions of the acquiring system - including the merchant account lifecycle, merchant discount rate, reconciliation, and settlement functionality.

Option 1: Acquirer System Configuration

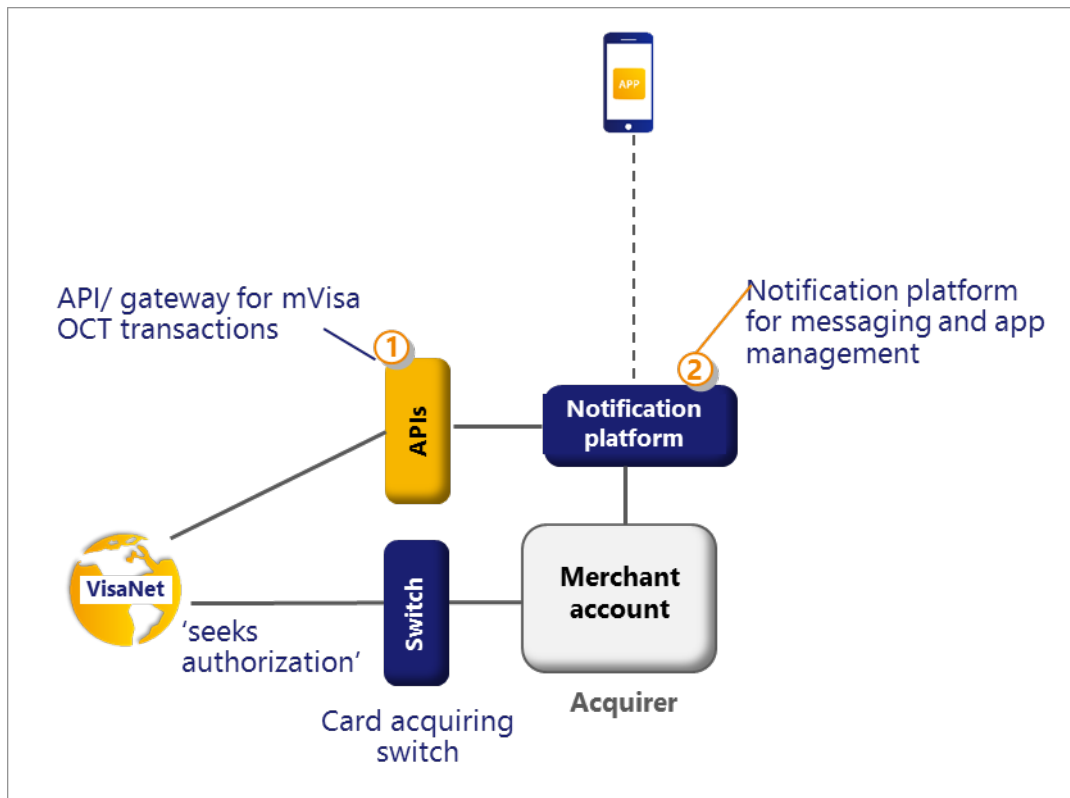


Configuration 2: Client using Visa API for OCT Transactions

In this option, the in-house financial switch is not used to process transactions.

Instead, APIs with linkages to VisaNet manage the transaction processing as depicted in the figure below. The acquirer has to modify (or create) the notification platform and put new processes in place to continue using functions of the acquiring system.

Option 2: Acquirer System Configuration



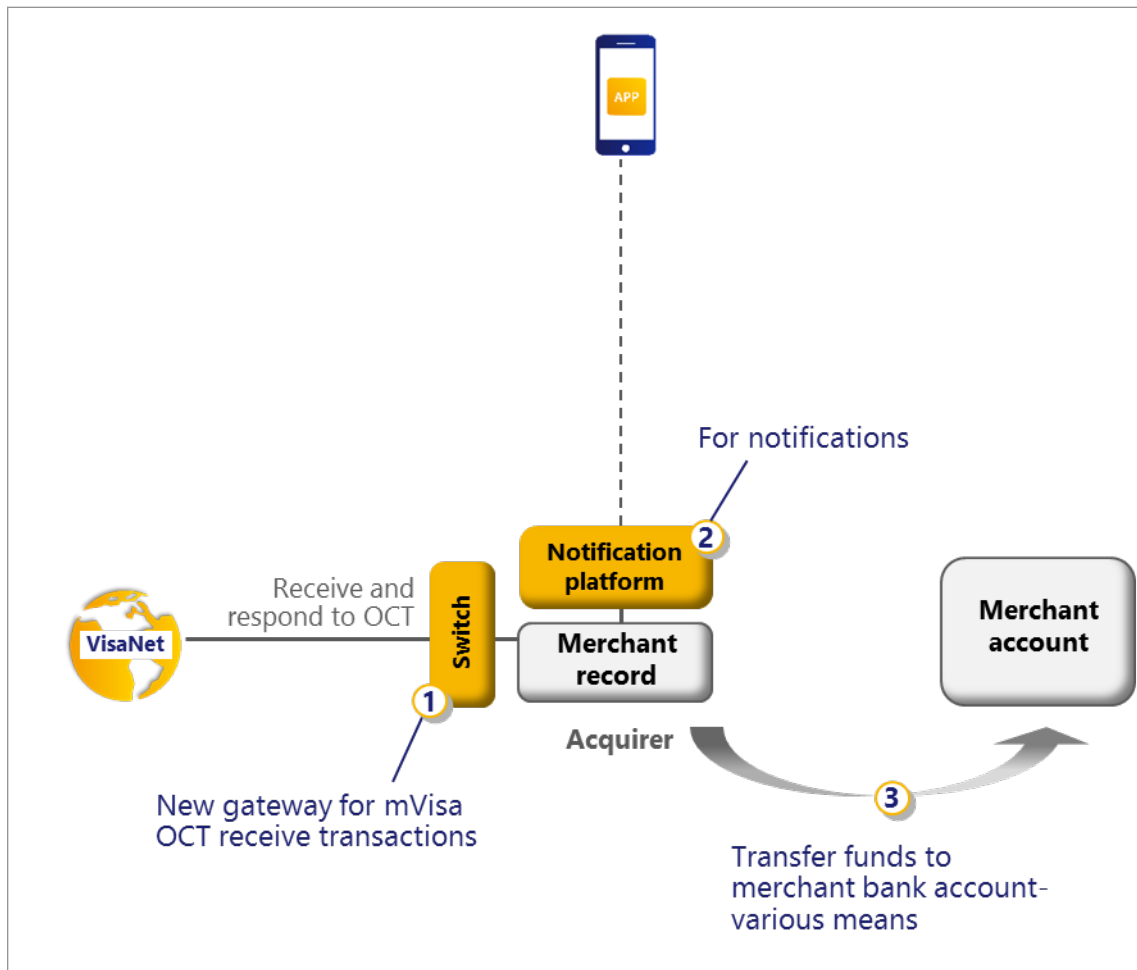
Configuration 3: Acquirer Using a Third Party Processing Service

In this option, the acquirer outsources the entire acquiring process as shown in the figure below.

The acquirer will need to outsource the following areas:

1. Switch for VisaNet transaction processing.
2. Notification platform for transactions.
3. Transfer funds to merchant bank accounts through various means.

Option 3: Acquirer System Configuration

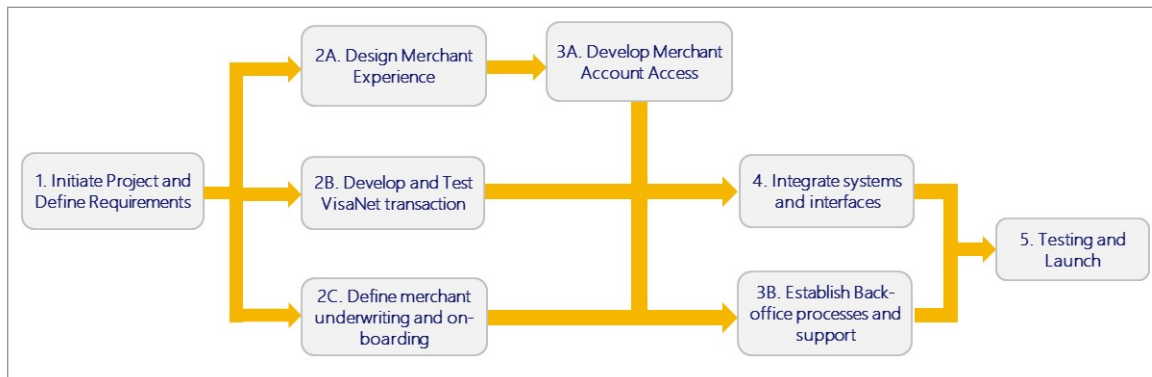


Acquirer Implementation Checklist

This section provides indicative implementation checklist for the acquirer.

It is for informational and reference purposes only, and should not be relied upon for operational, legal, technical, financial, or other advice. The checklist has been divided into typical stages of the acquirer implementation project. Typical stages in the acquirer implementation project for deploying the solution are depicted in the figure below.

Acquirer Implementation Project - Typical Stages



| Stage 1: Initiate Project and Define Requirements

Project Initiation

- Obtain executive sponsorship for the project.
- Identify project leader and project manager.
- Constitute core project team that has representation from different internal departments.
 - Seek regulatory approval as required.

Create and Publish Project Plan

- Conduct workshop with Visa to understand detail requirement.
- Refer Acquirer implementation overview document from Visa.
- Decide on the optional transactions to be included.
- Create a project implementation plan and agree on launch timelines with involved departments.
- Decide the type of accounts that will be included in the program.
- Evaluate phased versus big bang roll out.

Define Business Requirement

- Create a business requirement document (BRD).
- Refer Visa provide business requirement document template.
- Validate if the BRD is compliant to program requirement.
- Submit Program Registration form

Define Technical Requirement

- Determine high level system requirements
- Conduct gap analysis against existing capabilities.
- Evaluate mobile interface option – SDK versus app development.

- Compare effort required for API connection with ISO connection to Visa.
- Decide whether the system will be in-house or outsourced.
- If outsourced, conduct partner evaluation and selection.
- Finalize processing system configuration details.
- Seek budgetary approval.
- Issue purchase order to outsourced partner.
- Initiate project with Visa Implementation Manager.

| Stage 2A: Design Merchant Experience

Design Merchant Information Display

- Determine merchant account to merchant PAN one to many relationships.
- Plan merchant information display appropriate to merchant environment at point of transaction.
- Evaluate options to enable merchant information display at remote merchant.
- Identify merchant requiring additional data.
- Provide options for dynamic QR code.

Determine Notification method

- Identify risk control such as sales balance for the day, Bank as sender name in SMS notification.
- Enable in-app notification on merchant app in addition to SMS notification.
- Plan for server notification or cash register integration.
- Publish specification for easy server integration by merchants.
- Evaluate process for enabling QR code generation and notification print on POS terminals.
- Determine if a segment of merchant would be interested in email notification.
- All notification should contain the following data:
 - Amount
 - Additional amount
 - Additional data
 - Consumer Name
 - Masked Consumer PAN

One-to-many scenarios

- Merchant may require one merchant PAN resulting into more than one notification.
- Evaluate how combination of notification type, such as SMS as well as in-app, would be enabled for a given merchant PAN
- In case of large merchants, additional data such as bill number and/or reference number may be used to direct notification to a transaction point.

| Stage 3A: Develop Merchant Account Access

Access Options for the Merchant Account

- Evaluate the need for USSD, app or web access for merchant
- Enable sales history, merchant return, transaction logs and account details.
- Enable different user rights to a merchant account, depending on the role in merchant location e.g.:
 - owner would have complete account access,
 - store supervisor would process refund and access sales data, whereas,
 - frontline staff can only review notification

Merchant App Experience

- Determine and publish app support on various mobile operating systems
- Conduct session with Visa design team to identify most appropriate user experience.
- Plan to include different use case scenario such as tips, convenience fee etc.
- Submit wireframe of experience to Visa before starting development.
- Notification should as per business requirement.
- Share notification content with Visa.
- Develop mobile interface, USSD or app.
- Validate if app is capable of generating QR code as per requirement.
- Validate QR code generated by app using Visa test app.
- Submit user experience to Visa for approval.
- Establish and sign off mobile channel security requirement.
- Implement mobile channel authentication policy.
- Finalize and test mobile interface with backend.
- Enable additional services such as loyalty program, business intelligence and short-term working capital loan.

| Stage 2B: Develop and Test VisaNet Transaction

Submit Project Form to Visa

- Submit form to Visa Implementation Manager.
- Apply for a new BIN for assigning PAN to merchants.
- Obtain detailed project plan from implementation manager.
- Determine the VCMS test environment availability date.

Develop Interface between Systems

- Finalize specs for app to mobile processing system interface.
- Integrate and test app to mobile processing system interface.
- Develop interface between mobile channel backend system and VisaNet gateway.
- Plan for exception scenarios in transaction processing.
- Develop mobile processing system and merchant account interface.
- Conduct System Integration Testing.

Test VisaNet transaction

- Develop VisaNet messaging capability.
- Develop processing capability for merchant refund.
- Refer Visa test script and prepare for testing.
- Prepare for end to end testing starting from mobile interface to VisaNet.
- Conduct self (offline) testing with VCMS.
- Conduct attended (online) testing with Visa.
- Communicate production cut over steps and timelines to Visa.

Test QR code

- Generate QR code as per specification of Visa.
- Use Visa provided app for validation of QR code.

| Stage 2C: Define Merchant Underwriting and On-boarding

Merchant onboarding and maintenance

- Develop the merchant value proposition.
- Determine and document new merchant acquisition channel and approval process.
- Evaluate merchant sourcing process to include digital channel.
- Determine if remote processes can be utilized for merchant screening.
- Evaluate tiered screening and documentation requirement.

- Enable remote channel for sales and operations support.
- Evaluate changes to merchant transaction risk monitoring.
- Determine changes to merchant MIS.

Underwriting Policy

- Modify credit underwriting policy.
- Evaluate pre-approval for certain segment of merchant.

Determine merchant discount rate, commission, and limits

- Establish and approve merchant discount rate pricing policy.
- Merchant discount rate should be aligned to segment rates.
- Determine transaction limits.
- Establish criteria for instant credit versus deferred credit for merchant settlement.
- Determine commission to be paid out to merchant for cash transactions, if enabled.

| Stage 4: Integrate Processing Systems and Interfaces

Develop Integration between Systems

- Develop interfaces for the merchant information display
- Develop integration between notification platform and VisaNet transaction processing gateway
- Develop interfaces for merchant billing and settlements

| Stage 3B: Establish Back-office Processes and Support

Back Office Process and Support

- Establish process for printing of merchant information panel.
- Obtain sample approvals from Visa for merchant information panel.
- Establish a process of dispute handling.
 - Create dispute-handling document.
 - Obtain access to Visa Resolve Online (VROL) and identify internal processes.
- Determine the various reports that would be required.
 - Create regular report list and frequency.
 - Finalize procedure for report delivery and analysis.

- Determine call center support required for merchants.
 - Develop policies and procedures.
 - Identify point of contact for consumer queries.
 - Create quality standards.
 - Develop procedures for service enquiries.
 - Create customer service scripts.
 - Create welcome calling scripts and plan processes to make it operational.
- Identify settlement and reconciliation requirement and make back office system changes.
- Work with Finance and Accounting department to determine different ledger and income accounts.

Audit, Fraud and Risk Management

- Determine system controls and business processes to monitor fraud levels and manage the overall program risk.
- Conduct system IT security testing and penetration testing.
- Conduct technology risk assessment and establish controls.
- Determine and document ongoing Audit requirements and control for program.

Training

- Identify training needs and timelines for core project team, back office staff, call center staff and Branch staff.
- Develop training objectives and requirements as per audience.
- Develop training materials.
- Produce operational manuals and help guides.
- Produce internal departmental training schedule.

Legal Activities

- Finalize legal contracts and get internal legal approval.
- Sign all contracts or publish approved contracts.
- Obtain regulatory approval as required under prevailing regulation, if any.

| Stage 5: Testing and Launch

User Acceptance Testing

- Processing system testing.
- Internal user acceptance testing.

- Staff testing.
- Friends and family testing.

Finalize the Marketing Plan and Launch Planning

- Launch service backed with marketing activities.
- Develop a high-level marketing plan in accordance to the objectives and business plan.
- Develop merchant education plan.
- Define the public relations strategy for launch announcement.
- Define promotional and advertising strategies.
- Appoint and brief promotional and advertising agencies, if required.
- Ensure targeting and message content is consistent with strategic objectives.
- Design sales and training materials.
- Obtain legal approval for materials.
- Produce and distribute promotional materials.

Appendix A

Transaction Descriptions

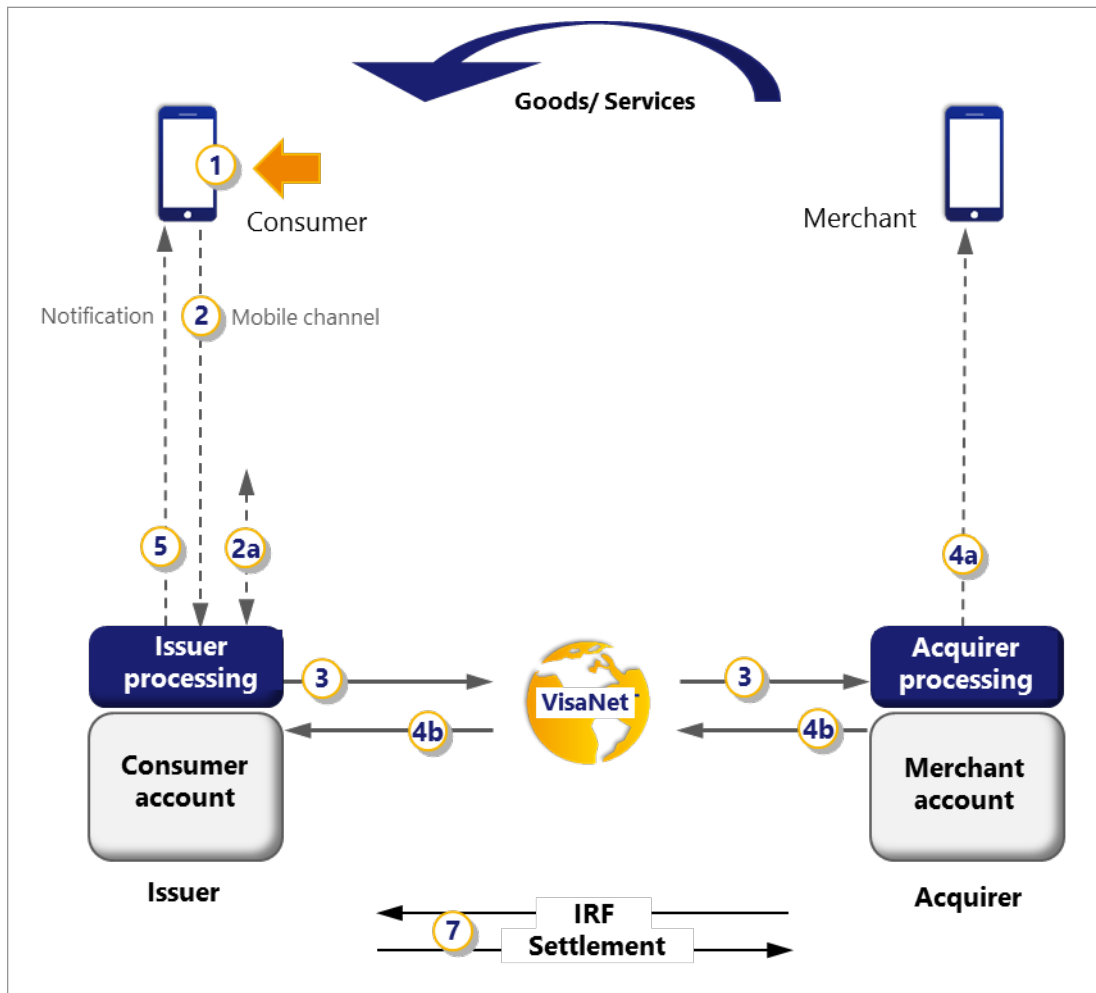
This section provides a step-by-step flow description of transactions defined in the program.

Merchant Payment Transaction

The consumer initiates the transaction using their mobile device by capturing the information from the merchant display.

The consumer and the merchant each receive a notification on their respective devices confirming the transaction. The merchant then completes the transaction.

Merchant Payment Transaction Flow in Steps



Merchant Payment Transaction Steps Described

Step	Location	Description
1	Physical and then on mobile device.	- Consumer makes a purchase of goods or services at a merchant. - Consumer initiates the merchant payment transaction via mobile device and captures the merchant information through any of the available option.
2	Over mobile channel.	Consumer instructions reach the issuer processing system via the mobile channel.
2a	Issuer processing.	If the merchant is internationally located, then the issuer processing system prompts the consumer to enter amount in international currency and requests the consumer to accept the forex conversion rate.
3	Issuer processing and then VisaNet.	- Issuer processing system authenticates the Consumer, determines the merchant PAN, and then performs Luhn check. - If both are successful, then the system confirms the amount in the consumer account and initiates SMS OCT to VisaNet (0200), which is routed to Acquirer by VisaNet.

Merchant Payment Transaction Steps Described

Step	Location	Description
4a	Merchant Acquirer processing.	- Acquiring processing system validates the merchant credentials and then for all validated transactions it records a payment favoring the merchant. The applicable merchant discount rate is calculated and applied. - Acquiring processing system also simultaneously triggers confirmation message to the merchant's device.
4b	Acquirer processing and then VisaNet.	The response message from the Acquirer (0210 or 0110) is routed back to Issuer over VisaNet.
5	Issuer processing.	Issuer processing system completes the transaction into the consumer's account and it also triggers a confirmation message to the consumer's mobile device.
6	Physical.	On receipt of the notifications, the merchant gives the consumer the goods or services and the transaction is complete.
7	VisaNet settlement.	VisaNet settles the transaction amount and interchange between the acquirer and issuer as per the applicable cut-off time.

In case of an international transaction, the following additional steps are required:

- 1. Identify Merchant Currency:** Issuer can identify the merchant currency using any of the following option (a) QR Code ID "53" transaction currency, or (b) Account Lookup API, or (c) Account Lookup File.
- 2. Determine Forex rate:** Based on the merchant currency, Issuer determines forex rate (including Issuer mark) between merchant currency and consumer currency, using Visa provided rates (available through TC 56 or API) or client's own internal rate table.
- 3. Amount in Merchant Currency:** Consumer, on app or USSD, is prompted to enter the amount in merchant currency and the final amount to be debited to consumer may be shown.

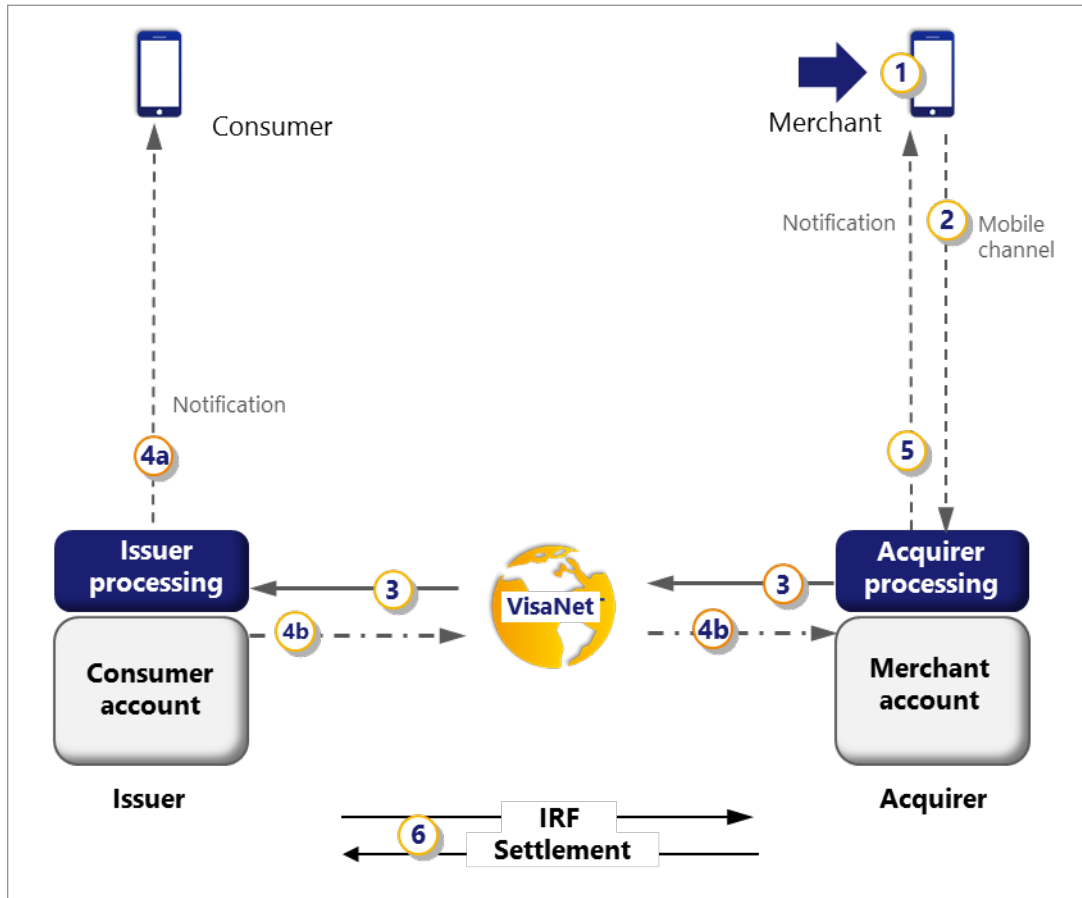
Refund or Cancellation of a Merchant Payment Transaction

A merchant may, at its discretion, process a credit into consumer account when a valid transaction was previously processed.

This situation can arise when the consumer cancels the purchase, or returns the goods in part or in full, or the merchant agrees to return a part of the payment.

The merchant will ask for the consumer PAN to initiate the transaction on the merchant device. The reversal can be processed by the merchant for the full amount (cancellation) or a partial amount (refund). The acquirer, depending on its setup may process the transaction online or via end of the day batch process. The consumer and the merchant each receive a notification on their respective devices confirming the cancellation or refund.

Refund and Cancellation of a Merchant Transaction Flow in Steps



Refund and Cancellation of a Merchant Transaction Steps Described

Step	Location	Description
1	Physical or remote and then on mobile device.	- Consumer returns a purchased good or cancels a paid-for service with the merchant who accepted payment using OCT at the time of original purchase. - Consumer provides their PAN number associated with initial purchase transaction to merchant, which is used via the merchant device to instruct full or partial amount to be returned to the consumer.
2	Over mobile channel.	Merchant instructions reach the acquirer processing system via the mobile channel or through other mechanism provided by the acquirer.
3	Acquirer processing and then VisaNet.	- Acquirer processing system authenticates the merchant and then performs Luhn check on consumer PAN. - If both are successful, then the system confirms the amount in the merchant account and initiates online (0200) or Base II (TC06) message for refund, which is routed to Issuer by VisaNet.
4a	Issuer processing.	Issuer processing system validates the consumer credentials, credits the amount into consumer account and simultaneously triggers confirmation message to the consumer's device.

Refund and Cancellation of a Merchant Transaction Steps Described

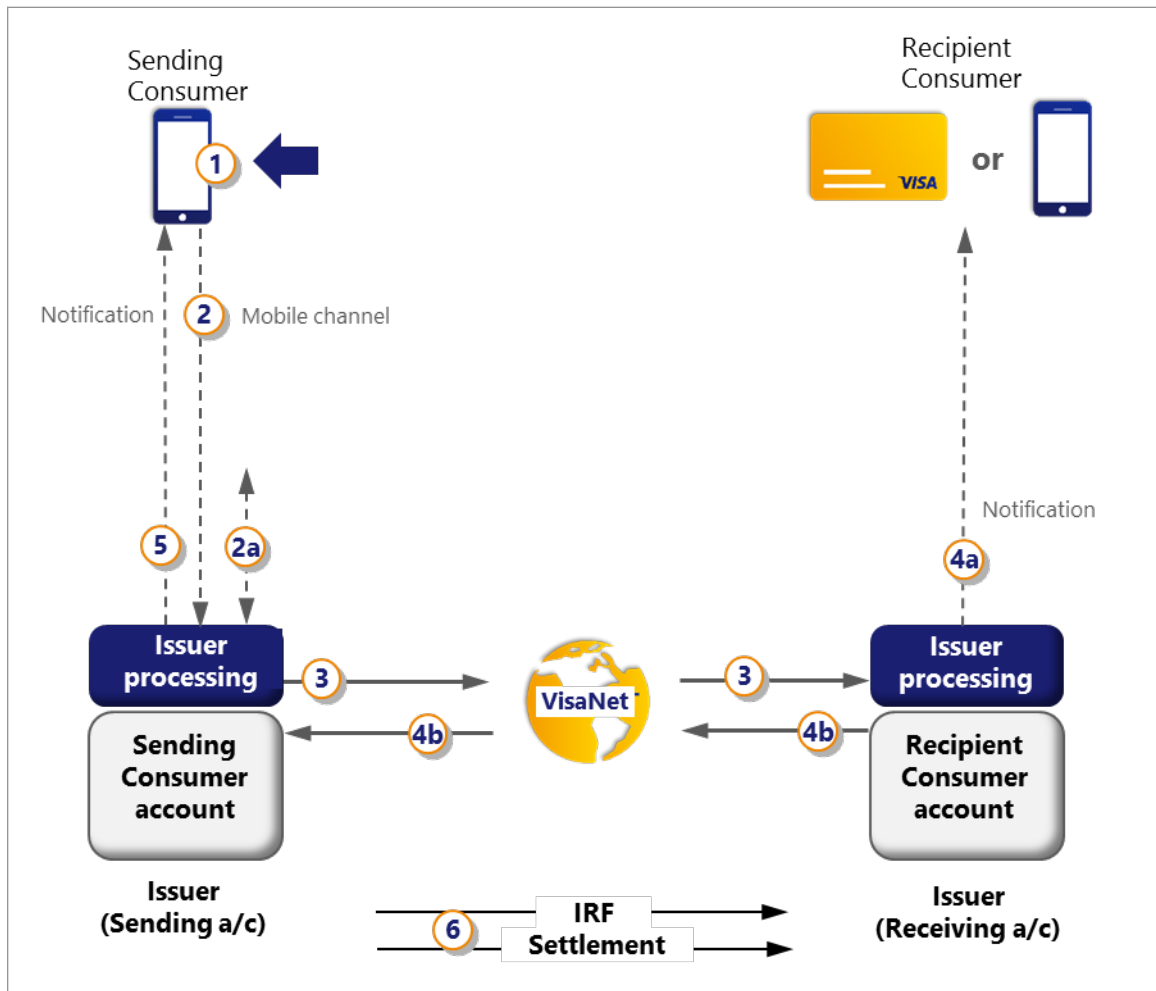
Step	Location	Description
4b	Issuer processing and then VisaNet.	In case of an online message from the acquirer, an online response is routed back to acquirer from the issuer processing system over VisaNet. This step would not take place in the case where the acquirer processes a Base II refund.
5	Issuer processing.	Acquirer processing system completes the transaction in merchant account and it also triggers confirmation message to the merchant mobile device. This step would not take place in case where the acquirer processes a Base II refund.
6	VisaNet settlement.	VisaNet settles the transaction amount and interchange between the acquirer and issuer as per the applicable cut-off time. ⁵⁶

Person-to-Person Money Transfer

A sender initiates the money transfer transaction using their mobile device and instructs transfer of the amount from their own account to the recipient account.

The recipient's PAN is used. The sender and the recipient each receive an SMS notification on their respective mobile devices confirming the transaction. Client should refer to *Visa Direct Original Credit Transaction (OCT) - Global Implementation Guide* for details.

Person-to-Person Money Transfer Transaction Flow in Steps



Person-to-Person Money Transfer Transaction Steps Described

Step	Location	Description
1	Physical and then on mobile device.	- Recipient consumer's 16-digit Visa PAN is communicated to the sender. - Sender initiates the person-to-person money transfer transaction via mobile device and provides the recipient's PAN.
2	On mobile device.	Consumer instructions reach the issuer processing system via the mobile channel.
2a	Sender Issuer processing.	If the recipient PAN has been issued in another country, the issuer processing system prompts the consumer to enter amount in international currency and requests the consumer to accept the forex conversion.
3	Sender Issuer processing and then VisaNet.	- Sender's issuer processing system authenticates the consumer and then performs Luhn check on recipient PAN. - If both are successful, then the system confirms the amount in the consumer account and initiates SMS OCT to VisaNet (0200), which is routed to recipient issuer processing system by VisaNet.

Person-to-Person Money Transfer Transaction Steps Described

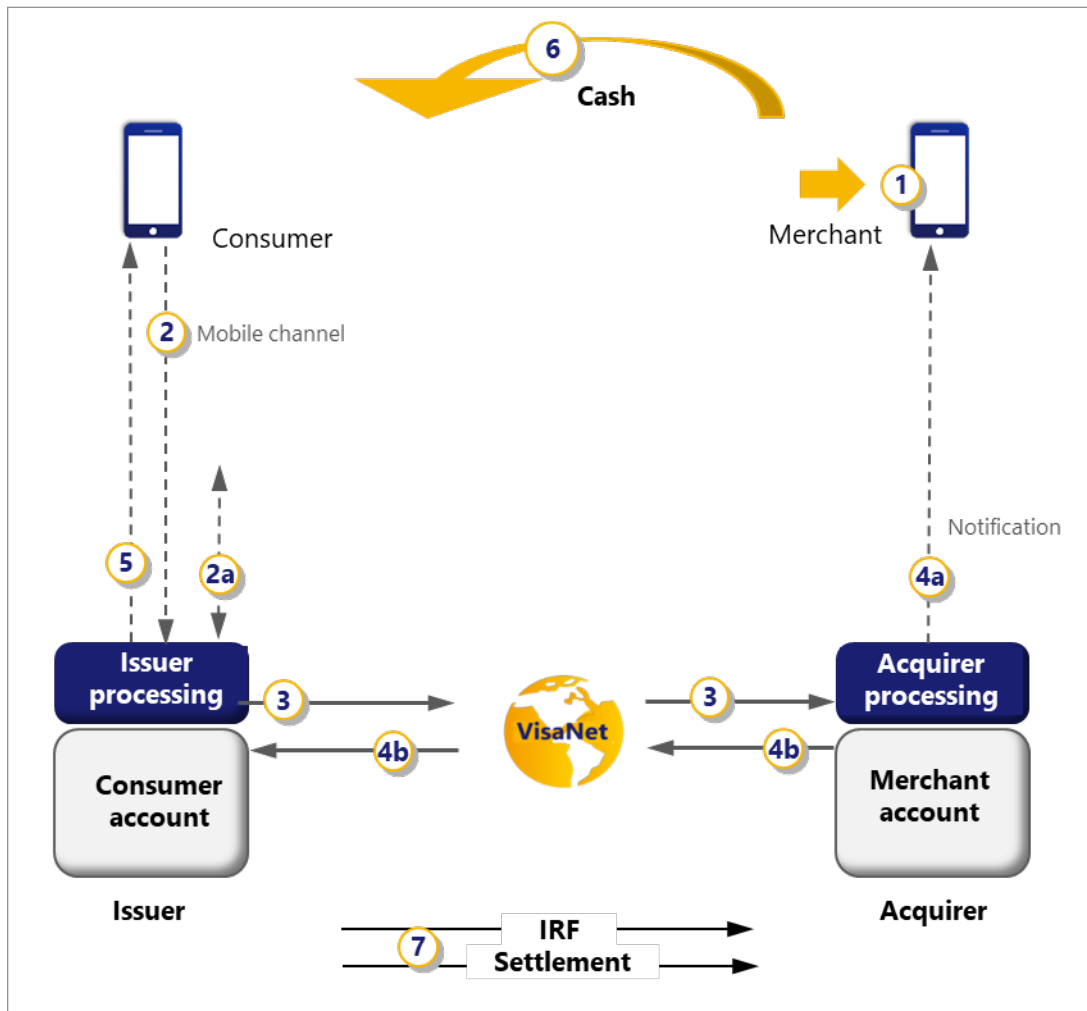
Step	Location	Description
4a	Recipient Issuer processing.	Recipient issuer processing system validates the PAN, credits the consumer account, and triggers a confirmation message to the recipient's mobile device.
4b	Recipient Issuer processing and then VisaNet.	The response message from the recipient processing system (0210 or 0110) is routed back to originating issuer processing system over VisaNet.
5	Sender Issuer processing.	Issuer processing system completes the transaction into the sender's account and it also triggers a confirmation message to the sender's mobile device.
6	VisaNet settlement.	VisaNet settles the transaction amount and interchange between the sender and recipient institution as per the applicable cut-off time.

Cash-Out (Withdrawal)

Consumers initiate the transaction using their mobile device and instruct transfer of the withdrawal amount from consumer account to the merchant account.

The consumer and the merchant each receive a notification on their respective mobile devices confirming the transaction. The merchant then gives an equal amount of cash to the consumer. The consumer signs the transaction record in the merchant log/receipt book.

Cash-Out (Withdrawal) Transaction Flow in Steps



Cash-Out (Withdrawal) Transaction Steps Described

Step	Location	Description
1	Physical	- Consumer approaches the merchant, (1) communicates the amount to be withdrawn, and (2) provides proof of ID. - Merchant provides confirmation that the required cash is available in their till for withdrawal. Merchant checks the consumer's proof of ID which is noted in the merchant log/receipt book.
2	On mobile device and then over mobile channel	- Consumer initiates the cash-out transaction via his mobile device and captures the merchant information through any of the available option. - Consumer instructions reach the issuer processing system via the mobile channel.
2a	Issuer processing	If the merchant is internationally located, then issuer processing system prompts the consumer to enter amount in international currency and requests the consumer to accept the forex conversion.

Cash-Out (Withdrawal) Transaction Steps Described

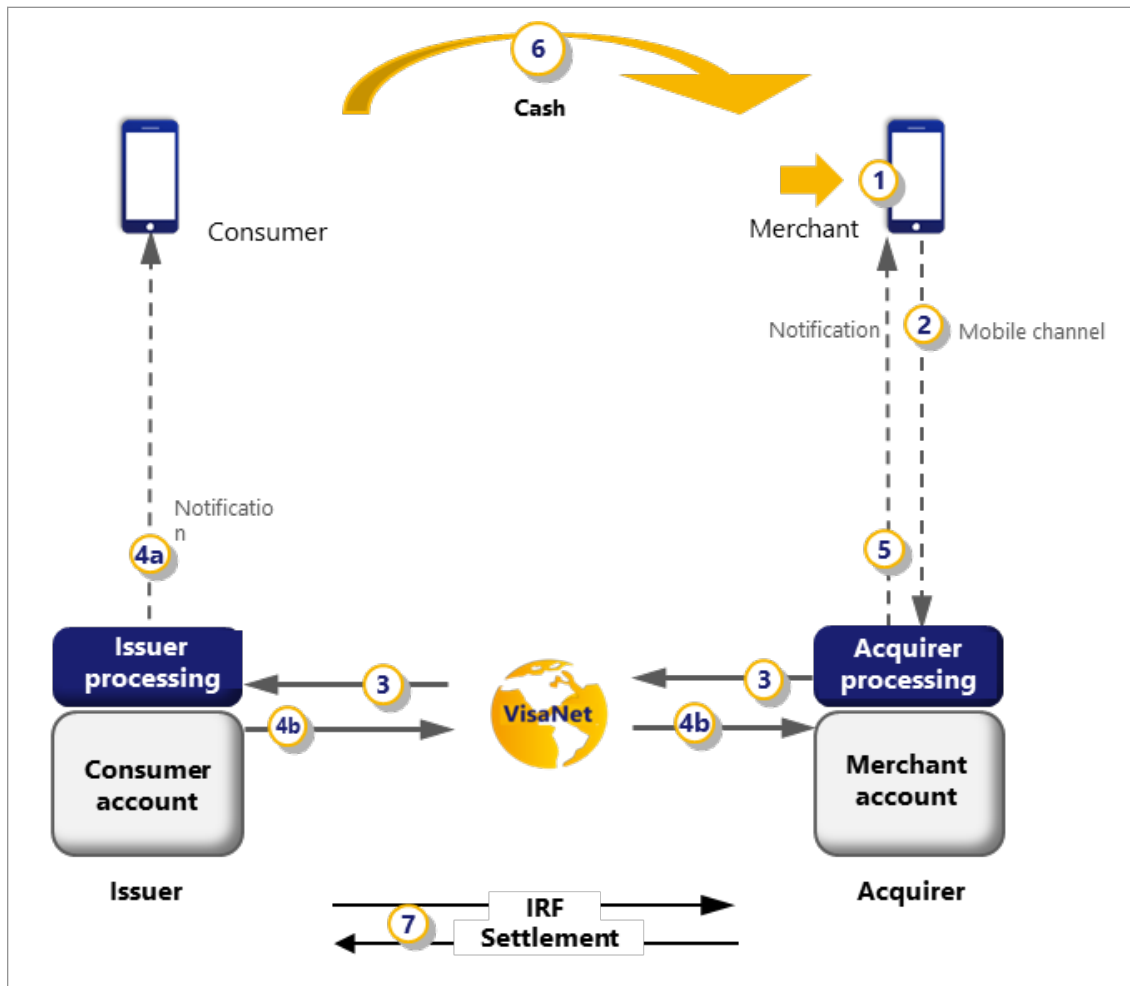
Step	Location	Description
3	Issuer processing	• Issuer processing system authenticates the consumer and then performs Luhn check on merchant PAN. • If both are successful, then the system confirms the amount in the consumer's account and initiates SMS OCT to VisaNet (0200), which is routed to the acquirer processing system by VisaNet.
4a	Acquirer processing	• Acquiring processing system validates the merchant credentials and then for all validated transactions it record's a payment favoring the merchant. The applicable merchant commission is calculated and applied. • Acquiring processing system also simultaneously triggers a confirmation message to the merchant's device.
4b	Acquirer processing and then VisaNet	The response message from the acquirer (0210 or 0110) is routed back to the issuer over VisaNet.
5	Issuer processing	Issuer processing system completes the transaction into the consumer's account and it also triggers a confirmation message to the consumer's mobile device.
6	Physical	On receipt of the SMS notifications, the merchant gives the cash to the consumer and the consumer signs the transaction record in the merchant log/receipt book.
7	VisaNet settlement	VisaNet settles the transaction amount and interchange between the issuer and acquirer as per the applicable cut-off time.

Cash-In (Deposit)

Merchants initiate the transaction using their mobile phone interface and instruct transfer of the deposit amount from their own (merchant) account to the consumer account.

The consumer and the merchant each receive an SMS notification on their respective mobile devices confirming the transaction. The consumer then gives the merchant the cash to be deposited and signs the transaction record in the merchant log/receipt book.

Cash-In (Deposit) Transaction Flow in Steps



Cash-In (Deposit) Transaction Flow Steps Described

Step	Location	Description
1	Physical and then mobile device	- Consumer approaches the merchant and (1) communicates the amount to be deposited, (2) provides proof of ID and (3) provides own 16-digit Visa PAN. - Merchant confirms availability of electronic funds in the merchant account. Merchant also checks the consumer's proof of ID and notes it in the merchant log/receipt book. - Merchant initiates the cash-in transaction via their mobile device by key entering the consumer's PAN.
2	Over mobile channel	Merchant instructions reach the mobile processing system of the acquirer via the mobile channel.

Cash-In (Deposit) Transaction Flow Steps Described

Step	Location	Description
3	Merchant Acquirer processing	• Acquirer processing system authenticates the merchant and then performs Luhn check on Consumer PAN. • If both are successful, then the system confirms the amount in the merchant account and initiates SMS OCT to VisaNet (0200), which is routed to issuer processing system by VisaNet.
4a	Issuer processing	• Issuer processing system validates the consumer credentials and then credits the consumer account. The applicable fees are also calculated and applied. • Issuer processing system also simultaneously triggers confirmation message to the consumer's device.
4b	Issuer processing and then VisaNet	The response message from the Issuer (0210 or 0110) is routed back to acquirer over VisaNet.
5	Acquirer processing	Acquirer processing system completes the transaction into the merchant account and it also triggers a confirmation message to the merchant's mobile device. The applicable merchant commission is also calculated.
6	Physical	On receipt of the SMS notifications, the consumer gives the cash to the merchant and the consumer signs the transaction record in the merchant log/receipt book.
7	VisaNet settlement	VisaNet settles the transaction amount and interchange between the issuer and acquirer as per the applicable cut-off time.

Appendix B

Key Terms and Acronyms Used

This section explains the acronyms and key terms used in this Mobile Push Payment Implementation Guide document.

Key Terms and Acronyms

Key Terms

Term	Description
Account	A financial store of monetary value with a providing entity that records transactions enabled using that store of value and also the resulting financial position of the account user with the providing entity. • Transactions on the Account are enabled for the solution – standalone method or in addition to transactions enabled using other means like payment card. • The Account can be held by a Consumer, a Merchant, or an Agent. • The Account can have various features and functionality, such as credit (loan or overdraft), savings, transactional or insurance.
Acquirer/Acquiring	Combined term used to refer to Merchant Acquirer and Agent Acquirer as defined in this table.

Key Terms

Term	Description
AML	Anti-Money Laundering (AML): A term used to describe those statutorily required controls that must be utilized by financial institutions to prevent their product or services from being used for money laundering activities.
ATM	Automatic Teller Machine
BAI	Business Application Identifier, the name of field indicator in the ISO8583 message
BIN	Bank Identification Number: The first 6 digits of the Payment card number that defines the issuer of that card.
CI	Cash In Transaction
CNP	Card Not Present Transaction
CO	Cash Out Transaction
Consumer	An individual or business entity who obtains financial services using the solution.
IRF	Interchange Reimbursement Fee
Issuer	Visa licensee that is the provider of Accounts to Consumers.
Interoperability	The ability for Visa issuers to transact through VisaNet with Visa acquirers without having to sign bilateral agreements.
KYC	Know Your Customer
GSM	Global System for Mobile Communications. The standards and system used by most countries for mobile cellular communications
Merchant	An individual or a business entity who accepts payment for goods, services using Mobile Push Payment, or facilitates deposit or withdrawal pursuant to a contract with a Merchant Acquirer.
Merchant acquirer	The Visa licensee that provides the Account to a Merchant as part of their contract to accept electronic payments using the solution for the sales of goods and services by the Merchant.
Structured message service (Mobile SMS)	The mobile GSM or CDMA standard text communication component of mobile phone service that delivers text-based event confirmation messages to mobile phones.
Mobile PIN	A numeric password entered on the mobile device and used by the consumer, issuer, agent acquirer or merchant acquirer to authenticate the account holder in the mobile channel.
MNO	Mobile Network Operator.
Mobile app or app	A computer program designed to run on smartphones, tablet computers and other mobile devices. Apps are usually available through application distribution platforms, typically operated by the owner of the mobile operating system.
Mobile SMS	Structured Message Service
Mod-10 or check digit	Redundancy check used for error detection of card numbers. It consists of a single digit computed from the other digits in the message. The computation method followed is the Luhn Algorithm.

Key Terms

Term	Description
MSISDN	Mobile Systems International Subscriber Identity Number a.k.a. Mobile number
National Net Settlement Service (NNSS)	Local in-country settlement system established by Visa to facilitate local rules and settlement in local currency.
OCT	Original Credit Transaction
Mobile Processing or Processing system	The software systems and the related processes utilized by the consumer issuer and the merchant acquirer to deliver mobile device access and push payment capability to the accounts provided by them. Mobile processing may be managed in-house by the account provider or can be outsourced to a third party.
Passcode	A numeric password protecting the PAN.
P&L	Profit and Loss Statement
PAN	Primary Account Number: The number on the Card following the industry standard for bank card numbers - ISO/IEC 7812
PCI-DSS	Payment Card Industry – Data Security Standards
PIN	Personal Identification Number
PRF	Program Registration Form for participation in the program
POS	Point-of-Sale
PSP	Payment Service Provider: New Regulatory category created in many countries under which non-bank institutions like mobile operators, retailers and NBFCs are able to provide payment services. The PSP regulatory license is less stringent and different from the 'Deposit taking' license for banks, but does not allow the PSP to perform financial intermediation and take 'deposits' from citizens.
User(s)	Consumer or merchant using the solution.
USSD	Unstructured Supplementary Service Data: GSM mobile standard that provides session-based communication. Is used by the mobile network to send information (usually text menus) between a mobile phone and an application on the network. Allows the subscriber to request information in short codes (starting with * and ending with #), or menus from the network via their mobile phone.
Single Message System (Visa SMS)	The ISO 8583 standard, full financial message that contains data for Authorization as well as Clearing and Settlement.
Visa Number or Visa PAN	The Visa-defined, 16-digit personal account number linked to a consumer account – it can be in virtual form or on the Visa card.
VROL	Visa Resolve Online
VSS	Visa Settlement Service